When Does Value-Weighted Targeting Pay? Budget-Constrained Retention Targeting in Online Gaming

Mark Stent*
Allwyn
mark@markstent.co.za

Abstract

A marketing team, a clinician, or the operator of a public-benefit programme often must pick a small fraction of a large population to treat, under a fixed budget, when people respond to treatment differently and are worth different amounts if they respond. Practitioners routinely rank candidates by the product of two numbers: an estimate of how much the treatment would change an individual's outcome, and an estimate of how much that improved outcome is worth (for instance, expected lifetime value). This paper places that heuristic inside the empirical-welfare-maximisation framework of Kitagawa and Tetenov (2018) and Athey and Wager (2021): we show the value-weighted product rule is the welfare-maximising budget-constrained policy, recovering the existing framework as the special case with unit values, and extend it to admit an operator-configurable targeting preference. This connects a decision rule already studied in marketing (Haupt and Lessmann, 2022; Hitsch and Misra, 2024) to the policy-learning literature and supplies the budget-constrained, preference-augmented version. Throughout, "welfare" denotes the operator's expected aggregate benefit (operator welfare), a quantity borrowed from econometrics; it is distinct from the wellbeing of the individuals targeted (player welfare), and the two can point in different directions.

Our central practical result is not that value weighting always helps, but a characterisation of *when each component earns its place*. A structural regret decomposition shows that welfare loss concentrates on individuals misranked near the targeting threshold, and attributes that loss to specific components of the score, telling an operator which estimate to improve first. Two operating regimes follow. First, value weighting helps most exactly when the treatment-effect model is weakest, because the value model retains information the treatment-effect model has dropped; it helps least at very tight budgets, where estimation variance in the value model dilutes the treatment-effect signal and ranking by treatment effect alone is preferable. Second, any operator preference that departs from neutral weakly reduces operator welfare, and we derive the exact size of that loss. This turns a governance question ("how restrictive should our targeting rules be?") into a price an operator can read off before deployment; the preference mechanism is therefore a tool for *pricing* restrictions, not for discovering welfare-improving ones, and we show calibration confirms this by construction.

We instantiate the framework for retention-bonus targeting in online gaming and study it in a controlled simulation. At realistic campaign budgets (reaching roughly 10% of the active base) ranking by treatment effect alone is competitive with or better than value weighting; the value-weighting advantage emerges at more generous budgets and under treatment-effect misspecification. A single joint-model estimator attains higher operator welfare than the decomposed rules in this DGP, and an end-to-end joint value-weighted model confirms the advantage is not an artefact of the comparison; a depth-matched ablation shows it is not a tree-depth artefact either. Several of our findings are therefore cautionary rather than promotional: the operator-preference weight provably cannot improve welfare, a plain joint model beats the decomposition an operator would otherwise reach for, and the decomposition's case rests on transparency and component-wise control rather than on welfare. The contribution we stand behind is the unifying welfare function that nests these comparisons,

*The views expressed are the author's own and do not necessarily represent the position of Allwyn.

the regret decomposition that says which estimate to improve first (we show it correctly identifies the binding component), and an honest map of when each component of the score does and does not pay. We also flag a domain hazard: because value can be high precisely when a player is gambling in harmful volumes, value-weighted targeting can concentrate bonuses on the most-harmed players, and we discuss the harm-protection safeguards a responsible deployment requires. The main limitation is the absence of real-data validation: the properties established here hold under controlled simulation (across two structurally different moderation shapes), and deployment performance depends on the quality of the component estimates in the wild.

Keywords: optimal treatment rules, causal inference, budget-constrained targeting, uplift modelling, policy learning, online gaming

1 Introduction

A decision-maker has a limited intervention (a retention bonus, a medical treatment, an educational programme) and a heterogeneous population. Each individual responds differently, is worth a different amount if they do respond, and has a different risk profile. The budget covers only a fraction of the population, and the decision-maker must choose which individuals to treat. This problem appears in personalised marketing (Lo, 2002; Hansotia and Rukstales, 2002), precision medicine, and public policy.

Some notation. Write X for the vector of pre-treatment covariates observed for each individual (age, spend history, and so on), taking values in the covariate space $\mathcal{X}$. Write $Y \in \{0, 1\}$ for the binary outcome of interest (e.g., whether the individual is retained next period) and $W \in \{0, 1\}$ for the treatment indicator (1 if the individual received the intervention, 0 otherwise). Under the standard potential-outcomes framework, $Y(1)$ denotes the outcome the individual would realise if treated and $Y(0)$ the outcome they would realise if not treated; only one of these is ever observed for any given individual, which is why treatment effects must be estimated rather than measured directly. A subscript i indexes individuals, so X_i, Y_i, W_i refer to the covariates, outcome, and treatment assignment of individual i. A hat denotes an estimate: $\hat{\tau}(x)$ is an estimate of $\tau(x)$, and likewise for other quantities introduced below.

The standard approach estimates the conditional average treatment effect (CATE)

$$\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x],$$

the expected gain in outcome from treating an individual with covariates x, and targets those with the highest estimates (Gutierrez and Gerardy, 2017; Künzel et al., 2019). Under a budget constraint the problem is one of optimal policy learning: choose a deterministic treatment rule $\pi : \mathcal{X} \to \{0, 1\}$ (a function that inputs an individual's covariates and outputs 1 if they should be treated and 0 otherwise) that maximises a *welfare function* $\mathcal{W}(\pi)$, subject to a cap $\kappa \in (0, 1)$ on the fraction of the population treated (Kitagawa and Tetenov, 2018; Athey and Wager, 2021). The welfare function is a scalar summary of how good a treatment rule is: loosely, the expected aggregate benefit produced when the rule is applied to the whole population, integrating over the distribution of covariates. In the canonical specification below, it is the expected treatment effect summed over the subset of the population the rule chooses to treat. The term is borrowed from econometrics and does not, on its own, refer to individual wellbeing. The canonical welfare function in this literature, empirical welfare maximisation (EWM), has two gaps that matter in practice. First, it treats every successful outcome as equally valuable: a low-value customer retained by treatment counts the same as a high-value regular. To fix this, this paper introduces an individual-level outcome value $V_i \geq 0$ (for example, customer lifetime value in a marketing setting, or quality-adjusted life-years in a medical one). Practitioners routinely multiply estimated uplift by customer lifetime value (LTV) to get a value-weighted ranking (Haupt and Lessmann, 2022; Gubela and Lessmann, 2021), but this heuristic has no

welfare-theoretic justification. Second, EWM gives the decision-maker no formal mechanism for encoding domain-specific targeting preferences. Such preferences are common. They reflect operational priorities, regulatory requirements, or responsible-practice considerations, not claims about causal mechanisms. To encode them, the paper introduces an auxiliary risk covariate $R_i \in [0, 1]$ (for example, the individual's churn probability) and a preference weight $g(R_i)$ that the operator configures. In online gaming, for instance, an operator may prefer not to waste bonuses on stable players, and may want to limit intervention on very high-churn players who may be disengaging for welfare-relevant reasons (Ascarza, 2018). There is currently no principled way to encode such preferences inside the policy-learning objective, and no way to quantify what they cost.

A small example to fix ideas. Consider eight players with different treatment effects, lifetime values, and churn risk scores (see Example 1). Ranking by pure treatment effect hands a bonus to the player with the largest predicted uplift even when that player is almost certain to stay without it. The budget is wasted on a sure thing. Multiplying treatment effect by lifetime value fixes this, but can still send a bonus to a very stable player if their value is large enough. A third factor, configured by the operator, lets them impose an explicit veto: do not send bonuses to players below a churn threshold, no matter what the other numbers say. This paper places the product rule on a welfare-theoretic footing (it is the welfare-maximising rule under a budget constraint), but its more useful contributions are diagnostic and conditional: a regret decomposition that tells the operator which of the three estimates is costing them welfare, a characterisation of when value weighting helps versus hurts, and a precise accounting of what the operator's preference factor costs when they choose to use it.

Contributions. We frame the paper deliberately as much around what does *not* pay as around what does: a unifying objective lets us state several cautionary results precisely (the operator-preference weight cannot improve welfare; a plain joint model beats the decomposition on welfare), and the positive contributions are a diagnostic and a map rather than a new optimal estimator.

1. **Welfare-theoretic placement of value-weighted targeting.** We show that ranking by estimated treatment effect times outcome value is the welfare-maximising rule under a budget constraint (Proposition 1), recovering the empirical-welfare-maximisation objective of Kitagawa and Tetenov (2018) and Athey and Wager (2021) when all values are equal. This connects the value-weighted decision rule studied in marketing (Haupt and Lessmann, 2022; Gubela and Lessmann, 2021; Hitsch and Misra, 2024) to budget-constrained policy learning. Neither value weighting (established by that prior work) nor budgeted ranking (established by the Qini/profit-uplift literature, Radcliffe, 2007; Devriendt et al., 2020) is itself new; our contribution is the threshold/shadow-price characterisation within empirical welfare maximisation, the regret decomposition below, and the operator preference extension.

2. **A regret decomposition that says which component to fix.** Welfare regret concentrates on individuals misranked near the targeting threshold (Proposition 2), and a telescoping identity attributes that regret to the estimation error in each component (Remark 8). This is the paper's main diagnostic: it tells an operator whether treatment-effect, value, or risk estimation is the binding constraint on welfare, and it explains the operating regimes below.

3. **When each component earns its place.** A simulation study identifies two regimes (Section 7): value weighting is most valuable when the treatment-effect model is misspecified (the value model recovers dropped signal) and least valuable at very tight budgets (where it dilutes the treatment-effect signal). The recommendation to use the product rule

is therefore budget- and model-quality-contingent, and we make the contingency explicit rather than leading with a single favourable budget.

4. **A tool for pricing targeting restrictions.** We add an operator-configurable preference weight on an auxiliary risk covariate and prove that any non-neutral setting weakly reduces operator welfare (Proposition 3). The practical output is a schedule of welfare costs for candidate restrictions, letting an operator price a governance or responsible-practice constraint before deploying it. We are explicit that welfare-based calibration of this weight returns the neutral setting by construction, so the mechanism prices preferences rather than discovering them.

We instantiate the framework for retention bonus targeting in online gaming, where the treatment effect is the causal impact of a bonus on retention, the outcome value is player lifetime value, and the preference weight encodes the operator's view on where along the churn spectrum to concentrate interventions.

Roadmap. Section 2 reviews related work. Section 3 fixes notation, identification assumptions, and a notation reference table. Section 4 states and proves the composite-score result. Section 5 gives the regret decomposition and the welfare-cost bound for preference weighting. Section 6 instantiates the framework for online gaming with a piecewise linear preference function. Section 7 reports the simulation study. Sections 8 and 9 discuss scope and conclude.

2 Related Work

2.1 Policy Learning and Empirical Welfare Maximisation

The optimal treatment rule literature seeks to assign treatments so as to maximise a population-level welfare criterion. Manski (2004) establishes minimax regret bounds (bounds on the worst-case welfare loss relative to an oracle rule) for treatment choice under ambiguity. Kitagawa and Tetenov (2018) formulate empirical welfare maximisation (EWM) with the welfare function $\mathcal{W}(\pi) = \mathbb{E}[\pi(X_i) \cdot \tau(X_i)]$, which is simply the expected treatment effect summed over the subpopulation that the rule π chooses to treat, and a budget constraint $\mathbb{E}[\pi(X_i)] \leq \kappa$ that caps the fraction of the population treated at κ. Under these conditions, the optimal deterministic rule assigns treatment to individuals whose CATE exceeds a threshold determined by the budget. Athey and Wager (2021) develop doubly robust policy learning with $O(n^{-1/2})$ asymptotic regret guarantees under complexity constraints on the policy class (for example, finite VC dimension), meaning that welfare regret shrinks at a rate proportional to $1/\sqrt{n}$ as the sample size n grows. The rate is conditional on those complexity constraints; flexible policy classes such as unrestricted gradient-boosted tree ensembles do not directly satisfy them, and the practical rate in such settings may be slower. Their framework supports budget, fairness, and simplicity constraints. Kallus (2018) introduces balanced policy evaluation, providing efficient off-policy estimators that improve upon inverse propensity weighting for policy comparison. Zhao et al. (2012) develop outcome-weighted learning for estimating individualised treatment rules. Zhou et al. (2023) extend offline policy learning to the multi-action setting with generalisation guarantees. Chernozhukov et al. (2018) provide generic machine learning inference methods for heterogeneous treatment effects. Neither Kitagawa and Tetenov (2018) nor Athey and Wager (2021) includes an individual-level outcome-value weight in the welfare function or a mechanism for covariate-dependent targeting preferences; their objectives are recovered from ours by setting $V_i = 1$ for every individual i and $g \equiv 1$. Our welfare function nests theirs as a special case and extends it along both dimensions.

2.2 Risk-Averse Policy Learning

Manski and Tetenov (2007) study treatment rules for a risk-averse social planner whose preferences are a concave function of aggregate welfare (that is, the planner's utility falls faster than welfare does at the margin, so they dislike variance in outcomes). Our formulation differs. The risk-modulation function $g(R_i)$, introduced in the preceding section, encodes preferences over an observable auxiliary covariate of the individual rather than over the distribution of aggregate welfare. This captures settings where the decision-maker has domain knowledge about which individuals are reachable by intervention, a common practical concern that existing risk-averse frameworks do not address.

2.3 Value-Weighted Uplift Targeting

Haupt and Lessmann (2022) study targeting under response-dependent costs and derive a profit-oriented decision rule in which candidates are ranked by a value- and cost-weighted treatment effect; ranking by $\hat{\tau}_i \cdot V_i$ (the product of estimated treatment effect and individual-level value) is the cost-homogeneous case. Here "uplift" is the marketing-literature term for the CATE, and V_i is the outcome value introduced above (typically customer lifetime value). Gubela and Lessmann (2021) propose value-driven evaluation metrics for uplift modelling that integrate treatment effect estimates with expected customer value, and the broader profit-uplift and profit-driven-analytics literature (Gubela et al., 2020; Fernández-Loría and Provost, 2022; Verbeke et al., 2017) frames targeting explicitly as expected-value (rather than effect) maximisation. Hitsch and Misra (2024) develop a decision-theoretic framework for heterogeneous targeting and characterise the optimal policy as ranking by expected incremental profit, i.e. effect weighted by margin or value. We do not claim to be the first to derive value-weighted targeting from a formal objective: this prior work already establishes its decision-theoretic merit. Our contribution is to place that rule inside the budget-constrained empirical-welfare-maximisation framework of Kitagawa and Tetenov (2018) and Athey and Wager (2021), where the optimal policy is a top-κ threshold rule (Proposition 1), and to extend it with the operator preference weight $g(R_i)$ and its welfare-cost accounting (Proposition 3).

The practice of allocating a fixed budget by ranking on an uplift-based score also has a long applied lineage in the Qini and profit-uplift literature (Radcliffe, 2007; Gubela et al., 2019; Devriendt et al., 2020), which evaluates budgeted targeting via cumulative-gain (Qini) curves and learns rankings directly for cost- or value-sensitive objectives. A budgeted top-κ cut of $\hat{\tau} \cdot \hat{V}$ is, operationally, a point on such a curve. We do not claim novelty for budgeted value ranking as a procedure; what the welfare-maximisation framing adds is (i) a precise optimality statement and threshold characterisation, (ii) the regret decomposition that attributes welfare loss to a specific component, and (iii) the preference weight g and its cost accounting. Devriendt et al. (2020) in particular formalise the observation, which our ordinal-sufficiency remark (Remark 7) also makes, that budgeted targeting depends only on the rank order of the score, not its calibrated magnitude.

2.4 Uplift Modelling

Uplift modelling provides the CATE estimates that enter our scoring rule. The framework is agnostic to the choice of CATE estimator. Any method that produces estimated individual treatment effects can serve as the uplift component, including T-Learners (which fit separate outcome models on the treated and control groups, then take their difference), S-Learners (which fit a single outcome model that takes treatment as an input feature), and X-Learners (which impute individual treatment effects before fitting a CATE model on those imputed values) (Künzel et al., 2019), causal forests (Wager and Athey, 2018), R-Learners (Nie and Wager, 2021), and doubly robust estimators (Kennedy, 2023).

2.5 Gaming Applications

Published applications of causal inference in online gaming remain limited relative to e-commerce and finance (Sifa et al., 2018; Braverman and Shaffer, 2012; Lemmens and Gupta, 2020); causal (as opposed to predictive) targeting in gambling specifically is close to absent, which is part of the motivation for this paper. The broader retention-management literature (Ascarza et al., 2018; Lemmens and Gupta, 2020) establishes that targeting the highest-risk customers is often ineffective and that responsiveness, not risk, should drive allocation, which is precisely the structure our risk-modulation function and the moderation $m(p)$ encode. Responsible-gaming considerations increasingly require that targeting algorithms exclude vulnerable players from promotional offers, introducing a domain-specific constraint not present in standard uplift formulations. Devriendt et al. (2018) provide a comprehensive survey of uplift modelling methods applicable in such settings. Our risk-modulation function provides a principled mechanism for encoding such constraints within the welfare-optimisation framework.

3 Setup and Notation

Consider a population of N individuals indexed by $i \in \{1, \ldots, N\}$, each characterised by a feature vector $X_i \in \mathcal{X} \subseteq \mathbb{R}^d$, where the covariate space $\mathcal{X}$ is a subset of d-dimensional Euclidean space and d is the number of covariates measured per individual. Let $W_i \in \{0, 1\}$ be the binary treatment indicator (1 if treated, 0 otherwise) and $Y_i \in \{0, 1\}$ the binary outcome (e.g., retention).

Assumption 1 (Causal identification)**.** *The following standard conditions hold: (i)* unconfoundedness*: the potential outcomes are independent of treatment assignment given the covariates,* $(Y_i(0), Y_i(1)) \perp W_i \mid X_i$ *(once we condition on* X_i*, the treatment assignment carries no further information about counterfactual outcomes); (ii)* bounded overlap*: there exists a constant* $\eta > 0$ *such that* $\eta \leq P(W_i = 1 \mid X_i = x) \leq 1 - \eta$ *for every* $x \in \mathcal{X}$*, so that every individual has a non-trivial probability of receiving either treatment assignment; (iii)* SUTVA *(stable unit treatment value assumption): no interference between units, so that treating one individual does not affect another's outcome; (iv)* consistency*:* $Y_i = Y_i(W_i)$*, so that the observed outcome coincides with the potential outcome under the treatment actually received.*

Assumption 2 (Regularity conditions)**.** *(i) The composite score* $s(X) = \tau(X) \cdot V(X) \cdot g(R(X))$ *has a continuous distribution conditional on* $g(R(X)) > 0$*. This ensures that the threshold* c^* *in Proposition 1 is unique and that ties between individuals occur with probability zero. Note that the risk-modulation function* g *can create atoms (point masses) in the score distribution when* $g(r) = 0$ *over a set of positive probability; the continuity condition only applies to the non-zero portion of the score support. A bounded-constant plateau in* g *(for example,* $g(r) = \phi > 0$ *on* $[p_H, 1]$ *in the piecewise linear specification of Section 6) does not by itself create atoms in* $s(X)$ *as long as* $\tau(X) \cdot V(X)$ *is continuously distributed conditional on that plateau, because the product* $\tau V \cdot \phi$ *inherits the continuous distribution of* τV*. (ii) The expectation* $\mathbb{E}[|\tau(X) \cdot V(X)|]$ *is finite, so that the welfare function has a well-defined value (averaging treatment-effect times value does not blow up to infinity).*

Under Assumption 1, the conditional average treatment effect is identified:

$$\tau(x) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x]. \tag{1}$$

In a randomised experiment where treatment is assigned independently of covariates, all four parts of Assumption 1 hold by design and $\tau(x)$ can be estimated from the observed data without further adjustment. In an observational setting (treatment assignment correlated with covariates) the same conditions still suffice in principle, but estimating $\tau(x)$ then requires modelling the

propensity score $P(W_i = 1 \mid X_i)$ or using a treatment-effect estimator that absorbs confounding internally (see Section 8.3 for the corresponding welfare-evaluation machinery).

In addition to the CATE, the decision-maker observes two auxiliary quantities for each individual:

Definition 1 (Auxiliary components)**.** *Let $V(x)$ denote the* outcome value *of individual i (e.g., lifetime value, quality-adjusted life-years), and let $R(x) \in [0, 1]$ denote the* risk covariate *(e.g., churn probability). Both are assumed to be pre-treatment quantities. We write $V(X_i) = V_i$ and $R(X_i) = R_i$ for notational convenience; $V(\cdot)$ and $R(\cdot)$ denote the underlying functions, while V_i and R_i denote their evaluation at individual i. We additionally assume $V(x) \geq 0$ for all x. This is a substantive restriction, not a normalisation: it rules out outcome-value metrics whose sign can flip for different individuals (for example, a harm-weighted welfare measure in a clinical trial where treatment is known to harm a subpopulation, so that V is negative for those individuals). The framework as stated applies to non-negative value metrics such as lifetime value or quality-adjusted life-years. Heterogeneous per-individual treatment* costs *are a separate concern, handled in Section 8 via the net-benefit score.*

Remark 1 (Conditional-mean reading of V and R)**.** *When V_i and R_i contain idiosyncratic noise beyond X_i (as in the simulated DGP of Section 7.1), $V(\cdot)$ and $R(\cdot)$ should be read as the conditional means $V(x) = \mathbb{E}[V_i \mid X_i = x]$ and $R(x) = \mathbb{E}[R_i \mid X_i = x]$. The welfare function* (3) *is invariant to this distinction by iterated expectations, and the oracle composite score throughout uses these conditional-mean quantities.*

Remark 2 (Pre-treatment requirement)**.** *If V_i or R_i reflects the effect of prior treatments (e.g., LTV elevated by previous promotional activity, churn risk modified by past interventions), conditioning on them introduces post-treatment bias and the causal interpretation of the composite score breaks down. This requirement constrains the auxiliary models to use only pre-treatment data.*

In operational practice the requirement is harder to satisfy than it looks. Customer lifetime value and churn probability in real systems are typically estimated from behavioural histories that were themselves shaped by prior treatments: spend patterns after previous bonuses, churn signals after previous campaigns, and so on. Strict compliance therefore requires either (i) segmenting individuals into cohorts defined before any promotional activity began, or (ii) partitioning the feature set of the auxiliary models to exclude any behavioural signal that post-dates treatment exposure. Practitioners deploying this framework on top of existing LTV and churn infrastructure should not assume their current models slot in cleanly. Their models may well include features that violate the pre-treatment requirement, and a careful audit of feature provenance is a necessary step before deployment.

A *deterministic treatment rule* is a function $\pi : \mathcal{X} \to \{0, 1\}$ specifying which individuals receive treatment. The decision-maker faces a budget constraint:

Definition 2 (Budget-constrained targeting)**.** *Given budget proportion $\kappa \in (0, 1)$, the feasible set of treatment rules is*

$$\Pi_\kappa = \{\pi : \mathcal{X} \to \{0, 1\} \mid \mathbb{E}[\pi(X_i)] \leq \kappa\}. \tag{2}$$

Notation reference

Table 1 collects the symbols used throughout the paper. Section numbers indicate where each quantity is defined.

Table 1: Notation reference.

Symbol	Meaning	First used
$X_i \in \mathcal{X}$	pre-treatment covariate vector for individual i	§3
$W_i \in \{0,1\}$	binary treatment indicator (subscripted; distinct from $\mathcal{W}(\pi)$)	§3
$Y_i \in \{0,1\}$	binary outcome (e.g., retention)	§3
$\tau(x)$	conditional average treatment effect (CATE)	§3
$V(x) \geq 0$	outcome value (e.g., lifetime value)	§3
$R(x) \in [0,1]$	auxiliary risk covariate (e.g., churn probability)	§3
$\pi : \mathcal{X} \to \{0,1\}$	deterministic treatment rule	§3
$\kappa \in (0,1)$	budget (fraction of population that may be treated)	§3
Π_κ	feasible set of treatment rules under budget κ	§3
$g : [0,1] \to [0,1]$	operator-configurable risk-modulation weight on R	§4
$\mathcal{W}(\pi; g)$	welfare of rule π: expected aggregate benefit, under preference g	§4
$s(x) = \tau(x)V(x)g(R(x))$	composite ranking score	§4
$h : \mathcal{X} \to [0,1]$	harm-protection weight on pre-treatment X	§6.2
$s^\dagger(x)$	harm-protected four-factor score, $\tau V g h$	§6.2
$s^\ddagger(x)$	harm-constrained penalised score, $\tau V g - \mu\,\text{harm}$	§6.2
c^*	welfare-optimal score threshold, satisfying the budget	§4
π^*	oracle-optimal rule; $\hat{\pi}$ its estimated counterpart	§4
$\mathcal{R}$	welfare regret of the estimated rule	§5
$\mathcal{M}$	misclassified set $\{i : \pi^*(X_i) \neq \hat{\pi}(X_i)\}$	§5
$\rho(\pi)$	risk exposure, mean R_i among treated	§5
$\theta = (p_L, p_{ML}, p_{MH}, p_H, \phi)$	parameters of the piecewise linear g	§6

4 The Operator's Decision Problem

4.1 Welfare Function

We define the operator's objective as a generalised welfare function. Recall that a welfare function $\mathcal{W}(\pi)$ in this literature is a scalar summary of the expected aggregate benefit of applying the rule π to the population. The canonical welfare function of Kitagawa and Tetenov (2018) sums treatment effects over the individuals the rule chooses to treat. The generalised version here weights each individual's treatment effect by two further quantities: the individual's outcome value (how valuable their response is) and an operator-configurable preference weight on an auxiliary risk covariate.

Definition 3 (Risk-modulated welfare)**.** *Let $g : [0,1] \to [0,1]$ be a* risk-modulation function*: a rule that takes a risk score $r \in [0,1]$ and returns a weight between 0 and 1. We require g to be measurable, a mild mathematical regularity condition ensuring that expectations involving g are well-defined; any sensible real-world function (step functions, piecewise linear functions, smooth functions) satisfies it. The non-negativity of g is substantive: it ensures that g does not flip the sign of the composite score (so a positive expected benefit cannot be turned into a negative one by the preference weight). The upper bound of 1 is a normalisation. Multiplying g by any positive constant leaves the ranking of individuals unchanged, so we lose no generality by capping g at 1. The risk-modulated welfare of a treatment rule π is*

$$\mathcal{W}(\pi; g) = \mathbb{E}\big[\pi(X_i) \cdot \tau(X_i) \cdot V_i \cdot g(R_i)\big]. \tag{3}$$

The multiplicative form is a modelling assumption: it requires that the operator's valuation of treating individual i is separable into the three components. We discuss the scope and limitations of this assumption in Section 8. When $g \neq 1$, the operator is expressing a preference for concentrating treatment on specific risk segments, accepting a reduction in $\mathcal{W}(\pi; \mathbf{1})$ (the unmodulated welfare) in exchange for targeting properties they value. The function g thus plays a different role from τ and V: the latter two are estimable quantities about the world, while g encodes the operator's preferences about where to intervene.

The welfare function (3) has a direct interpretation: it is the expected total value generated by treatment, where each individual's contribution is their treatment effect (τ), weighted by how valuable their response is (V), and further modulated by the operator's preference weight for their risk profile (g).

Remark 3 (Nesting existing frameworks)**.** *Setting $V_i = 1$ and $g \equiv 1$ recovers the standard EWM welfare of Kitagawa and Tetenov (2018): $\mathcal{W}(\pi) = \mathbb{E}[\pi(X_i) \cdot \tau(X_i)]$. Setting $g \equiv 1$ alone recovers value-weighted welfare: $\mathcal{W}(\pi) = \mathbb{E}[\pi(X_i) \cdot \tau(X_i) \cdot V_i]$, the implicit objective underlying the empirical practice of Haupt and Lessmann (2022) and Gubela and Lessmann (2021). The full formulation introduces the risk-modulation function g as the novel component.*

4.2 Optimal Treatment Rule

Proposition 1 (Optimal risk-modulated treatment rule)**.** *Under Assumptions 1 and 2, the treatment rule that maximises $\mathcal{W}(\pi; g)$ subject to $\pi \in \Pi_\kappa$ is*

$$\pi^*(x) = \mathbf{1}\{\tau(x) \cdot V(x) \cdot g(R(x)) > c^*\}, \tag{4}$$

where c^* is the unique threshold satisfying $\mathbb{E}[\pi^*(X_i)] = \kappa$. Equivalently, π^* targets the top-κ fraction of individuals ranked by the composite score

$$s(x) = \tau(x) \cdot V(x) \cdot g(R(x)). \tag{5}$$

Proof. The problem is to choose the rule π that maximises

$$\max_{\pi \in \Pi_\kappa} \mathbb{E}\big[\pi(X_i) \cdot \tau(X_i) \cdot V_i \cdot g(R_i)\big]$$

subject to the budget cap. The standard way to handle a budget constraint in an optimisation problem is the method of Lagrange multipliers: we attach a non-negative number $\lambda \geq 0$, called a *Lagrange multiplier*, to the budget constraint and add it to the objective. Doing so gives the *Lagrangian*

$$\mathcal{L}(\pi, \lambda) = \mathbb{E}\big[\pi(X_i)(\tau(X_i) \cdot V_i \cdot g(R_i) - \lambda)\big] + \lambda\kappa.$$

The multiplier λ is a bookkeeping device at this stage; we will see below that at its optimum value it coincides with the score threshold c^*, and Remark 4 gives the economic interpretation of that threshold as the shadow price of the budget. Reading the Lagrangian, for any fixed λ the decision-maker is free to set $\pi(x) \in \{0, 1\}$ individual by individual. Each individual's contribution to the Lagrangian is $\pi(x)$ times the quantity $\tau(x) \cdot V(x) \cdot g(R(x)) - \lambda$. Because $\pi(x)$ is binary, the individual-level contribution is maximised by setting $\pi(x) = 1$ whenever that quantity is positive and $\pi(x) = 0$ otherwise. This gives the threshold rule

$$\pi_\lambda(x) = \mathbf{1}\{\tau(x) \cdot V(x) \cdot g(R(x)) > \lambda\}.$$

(This kind of argument, where the maximiser is pinned down individual by individual rather than globally, is called a *pointwise* argument.) At the boundary $\tau(x)V(x)g(R(x)) = \lambda$ either choice of $\pi(x) \in \{0, 1\}$ is equally optimal for the Lagrangian; Assumption 2(i) makes this boundary set a measure-zero event, so the strict vs non-strict inequality in the threshold rule has no welfare consequence. Setting $c^* = \lambda^*$, where λ^* is chosen to exactly exhaust the budget ($\mathbb{E}[\pi_{\lambda^*}(X_i)] = \kappa$), yields the optimal rule. This characterisation presumes the budget binds, which requires that at least a fraction κ of the population has a strictly positive composite score: $P(s(X) > 0) \geq \kappa$. When this holds, the existence and uniqueness of $\lambda^* > 0$ (equivalently $c^* > 0$) follow from the continuity condition in Assumption 2. When it fails, that is, when fewer than a fraction κ of individuals have a positive score, the constraint slacks: by complementary slackness $\lambda^* = 0$, the optimal rule treats every individual with $s(X) > 0$ (and no one with $s(X) \leq 0$, since treating a

non-positive-score individual cannot raise welfare), and $c^* = 0$ is no longer unique on the atom at zero. This degenerate case is exactly what an aggressive preference weight with a wide $g \equiv 0$ exclusion zone can induce. The remaining step is to verify that solving the Lagrangian gives the solution to the original constrained problem, a property called *strong duality*. Strong duality holds for the *convex relaxation* of the problem, in which $\pi(x)$ is allowed to take any value in $[0, 1]$ rather than just $\{0, 1\}$: the relaxed problem has a linear objective and a linear constraint, and any rule that treats strictly fewer than a fraction κ of the population is strictly feasible, so Slater's condition applies. The relaxed optimum is attained by a binary-valued rule almost surely: for each x the pointwise-optimal $\pi(x) \in [0, 1]$ is 1 when the integrand is positive and 0 when it is negative, and the measure-zero boundary noted above makes any intermediate value $\pi(x) \in (0, 1)$ irrelevant. The relaxed solution therefore coincides with the solution to the original binary problem. The overall argument (a threshold rule on an objective integrand maximises the objective under a single budget constraint) is the standard Neyman-Pearson logic used in statistical testing and selection problems, applied by Kitagawa and Tetenov (2018, Theorem 1) to the CATE-only objective and extended here to the three-factor product $\tau(x) \cdot V(x) \cdot g(R(x))$. □

Remark 4 (Interpretation of the threshold)**.** *The threshold c^* has a simple economic meaning: it is the minimum composite score at which it is worth spending a unit of budget on an individual. Anyone with a score above c^* is treated; anyone below is not. In economic terminology, c^* is the* shadow price *of the budget: the welfare that a marginal unit of additional budget would buy. As the budget grows ($\kappa \to 1$), more individuals clear the bar, so c^* falls toward the lowest composite score present in the population (the* infimum *of the score distribution). In the unconstrained limit, every individual with a positive composite score is treated.*

Remark 5 (What g encodes)**.** *The function g is a mechanism for the operator to express a preference about where along the auxiliary-covariate distribution they want to intervene. It enters the welfare function multiplicatively, which has three consequences. First, setting $g(r) = 0$ for some values of r places zero weight on treating individuals with that risk profile no matter what their treatment effect or value; this veto property lets the operator encode hard exclusions (e.g., do not target very low-churn players) directly in the objective rather than as an external post-hoc filter. Second, because the welfare function is multiplicative, g is distinct in character from τ and V: the latter two are estimable quantities about the world; g is a preference, not a claim about causal mechanisms. Third, an operator who chooses $g \neq 1$ accepts a quantifiable welfare cost (Proposition 3) in exchange for the targeting pattern they prefer. Even a perfectly specified CATE model does not foreclose choosing $g \neq 1$; that choice just has to be justified on grounds other than welfare.*

4.3 Decomposability and Ordinal Sufficiency

Remark 6 (Decomposability)**.** *The proof is a standard Lagrangian argument, so the result's contribution is the decomposition rather than the optimality per se. The composite score $s(x) = \tau(x) \cdot V(x) \cdot g(R(x))$ separates the targeting decision into three components that can be estimated, validated, and updated independently. If the operator replaces any single component while holding the others fixed, the resulting score remains welfare-maximising for the correspondingly modified welfare function; this follows from the multiplicative structure of* (3)*. The operator can audit the uplift model, the value estimates, and the risk-modulation function separately, diagnose failures in any one of them, and swap in an improved version without retraining the others. A single-model policy learner does not have this property and must be retrained end-to-end when any aspect of the targeting objective changes.*

Remark 7 (Ordinal sufficiency)**.** *The top-κ targeting rule only cares about the rank ordering of individuals by composite score, not about the scores' exact magnitudes. (Ranking "by order" rather than by "size" is the sense in which the rule is* ordinal *rather than* cardinal*.) So if an*

estimated score $\hat{s}(x) = \hat{\tau}(x) \cdot \hat{V}(x) \cdot \hat{g}(\hat{R}(x))$ happens to rank individuals in the same order as the true (oracle) score $s(x)$, then the estimated rule picks exactly the same set of people as the oracle rule. Formally, $s(x) > s(x') \Leftrightarrow \hat{s}(x) > \hat{s}(x')$ for all pairs implies the top-κ sets coincide. Rank preservation is, however, a strong condition, especially for a product of three estimated components. Even if each component's estimator is monotone in the true component (for example, $\hat{\tau}(x) = h_\tau(\tau(x))$ for some strictly increasing h_τ), the product $h_\tau(\tau) \cdot h_V(V)$ may rank individuals differently from the true product $\tau \cdot V$ when h_τ and h_V are different transformations. A simple example: suppose $\hat{\tau} = \tau^2$ and $\hat{V} = V$. An individual with $(\tau = 1, V = 10)$ and an individual with $(\tau = 2, V = 3)$ have true products $1 \cdot 10 = 10$ and $2 \cdot 3 = 6$, so the first ranks higher. Their estimated products are $1^2 \cdot 10 = 10$ and $2^2 \cdot 3 = 12$, reversing the order. When ranks are not perfectly preserved, the welfare loss depends on how much oracle-score mass sits on the set of misranked individuals near the targeting threshold (Proposition 2 makes this precise).

5 Properties

5.1 Welfare Regret Decomposition

We now ask: if the operator uses estimates $(\hat{\tau}, \hat{V}, \hat{g})$ instead of the true (but unknown) quantities (τ, V, g), how much welfare does the resulting rule leave on the table? The gap between the oracle rule (which uses the truth) and the estimated rule is the *welfare regret*. Call the oracle rule π^* and the estimated rule $\hat{\pi}$, and call the regret $\mathcal{R} = \mathcal{W}(\pi^*; g) - \mathcal{W}(\hat{\pi}; g)$.

The result below is a *structural decomposition*: it rewrites the regret in a form that reveals where the loss comes from, rather than a *statistical convergence bound* that tells us how fast the regret shrinks as we collect more data. Both are useful; they answer different questions.

Proposition 2 (Welfare regret decomposition)**.** *Let π^* denote the oracle-optimal rule (which uses true τ, V, g) and $\hat{\pi}$ the estimated rule (which uses $\hat{\tau}$, $\hat{V}$, $\hat{g}$). Define the regret $\mathcal{R} = \mathcal{W}(\pi^*; g) - \mathcal{W}(\hat{\pi}; g)$. Then*

$$\mathcal{R} = \mathbb{E}\big[(\pi^*(X_i) - \hat{\pi}(X_i)) \cdot \tau(X_i) \cdot V_i \cdot g(R_i)\big] = \mathbb{E}\big[(\pi^*(X_i) - \hat{\pi}(X_i)) \cdot s(X_i)\big], \tag{6}$$

where $s(x) = \tau(x) \cdot V(x) \cdot g(R(x))$ is the oracle composite score. Intuitively: the regret only picks up contributions from individuals the two rules disagree on. Call this the misclassified set*: $\mathcal{M} = \{i : \pi^*(X_i) \neq \hat{\pi}(X_i)\}$. The regret is bounded by the average oracle score on that set:*

$$|\mathcal{R}| \leq \mathbb{E}\big[|s(X_i)| \cdot \mathbf{1}\{i \in \mathcal{M}\}\big]. \tag{7}$$

When both rules exactly exhaust the budget ($\mathbb{E}[\pi^*] = \mathbb{E}[\hat{\pi}] = \kappa$), each false negative ($\pi^* = 1$, $\hat{\pi} = 0$) is paired with a false positive ($\pi^* = 0$, $\hat{\pi} = 1$), and the c^* contributions cancel, giving the tighter bound

$$\mathcal{R} = \mathbb{E}\big[|s(X_i) - c^*| \cdot \mathbf{1}\{i \in \mathcal{M}\}\big], \tag{8}$$

which shows that misclassification hurts welfare in proportion to how far each misclassified score sits from the threshold, not in proportion to its absolute magnitude. In words: estimation mistakes only hurt welfare to the extent that the mistaken individuals are near the margin of being worth treating. The budget-exhaustion condition holds automatically for top-κ-percentile selection on a continuously distributed score, which is the rule used by π^* and by every estimated strategy in Section 7; if instead $\hat{\pi}$ is obtained by thresholding an estimated score and the threshold does not exhaust the budget, the looser bound (7) *still applies.*

Proof. By linearity of expectation,

$$\begin{aligned}\mathcal{R} &= \mathbb{E}\big[\pi^*(X_i) \cdot s(X_i)\big] - \mathbb{E}\big[\hat{\pi}(X_i) \cdot s(X_i)\big] \\ &= \mathbb{E}\big[(\pi^*(X_i) - \hat{\pi}(X_i)) \cdot s(X_i)\big].\end{aligned}$$

For any individual i not in $\mathcal{M}$, the two rules agree, so $\pi^*(X_i) - \hat{\pi}(X_i) = 0$ and that individual contributes nothing to the expectation. The regret therefore reduces to an expectation over the misclassified set alone. Taking absolute values and using $|\pi^* - \hat{\pi}| \leq 1$ (the difference of two binary indicators is between -1 and 1) gives (7). For (8), decompose $\mathcal{M}$ into false negatives $\mathcal{M}^+ = \{\pi^* = 1, \hat{\pi} = 0\}$ and false positives $\mathcal{M}^- = \{\pi^* = 0, \hat{\pi} = 1\}$, and write

$$\mathcal{R} = \mathbb{E}[s(X_i) \cdot \mathbf{1}\{\mathcal{M}^+\}] - \mathbb{E}[s(X_i) \cdot \mathbf{1}\{\mathcal{M}^-\}].$$

When both rules exhaust the budget, $\mathbb{E}[\pi^*] = \mathbb{E}[\hat{\pi}] = \kappa$, so $\mathbb{E}[\pi^* - \hat{\pi}] = 0$; since $\pi^* - \hat{\pi} = \mathbf{1}\{\mathcal{M}^+\} - \mathbf{1}\{\mathcal{M}^-\}$, this gives the equal-mass identity $\mathbb{E}[\mathbf{1}\{\mathcal{M}^+\}] = \mathbb{E}[\mathbf{1}\{\mathcal{M}^-\}]$ (equality of expected mass, not of cardinality). Consequently $c^* \cdot (\mathbb{E}[\mathbf{1}\{\mathcal{M}^+\}] - \mathbb{E}[\mathbf{1}\{\mathcal{M}^-\}]) = 0$, so subtracting and adding $c^* \cdot \mathbf{1}\{\mathcal{M}^\pm\}$ leaves the expectation unchanged and yields

$$\mathcal{R} = \mathbb{E}[(s(X_i) - c^*) \cdot \mathbf{1}\{\mathcal{M}^+\}] + \mathbb{E}[(c^* - s(X_i)) \cdot \mathbf{1}\{\mathcal{M}^-\}].$$

On $\mathcal{M}^+$ the oracle treats, so $s(X_i) \geq c^*$; on $\mathcal{M}^-$ it does not, so $s(X_i) \leq c^*$. Both terms are therefore non-negative and combine to give $\mathcal{R} = \mathbb{E}[|s(X_i) - c^*| \cdot \mathbf{1}\{i \in \mathcal{M}\}]$. □

Remark 8 (Component-wise score error)**.** *The bound* (7) *tells us that regret depends on the oracle-score magnitude on the misclassified set. A related question is* why *individuals end up misclassified, and what to do about it. The score estimation error for a single individual is* $|s(x) - \hat{s}(x)| = |\tau(x)V(x)g(R(x)) - \hat{\tau}(x)\hat{V}(x)\hat{g}(\hat{R}(x))|$. *Applying the* triangle inequality *(the elementary fact that* $|a + b| \leq |a| + |b|$*) after replacing oracle components with their estimates one at a time gives the bound*

$$|abc - \hat{a}\hat{b}\hat{c}| \leq |a - \hat{a}| \cdot |b| \cdot |c| \;+\; |\hat{a}| \cdot |b - \hat{b}| \cdot |c| \;+\; |\hat{a}| \cdot |\hat{b}| \cdot |c - \hat{c}|.$$

This manoeuvre is called a telescoping argument *(each replacement extends the "telescope" one section). The bound decomposes the total score error into three terms, one per component. It is diagnostic: it tells the operator which component's estimation error contributes most to misranking at the targeting threshold. Errors in components with large magnitude (for example, outcome value* V*, which can vary by orders of magnitude when* V *is customer lifetime value) are amplified by the magnitudes of the other two components. The risk-modulation function* g*, being bounded in* $[0, 1]$*, has limited scope to amplify errors in the other components.*

Remark 9 (Comparison with statistical regret bounds)**.** *Our regret decomposition answers a different question from the finite-sample regret bounds of Athey and Wager (2021). Athey and Wager establish that under their empirical welfare maximisation setup, welfare regret shrinks at rate* $O(n^{-1/2})$ *as sample size* n *grows (so doubling the sample roughly halves the square-rooted regret). That is a* statistical *convergence rate: how welfare regret shrinks with more data. Proposition 2 is* structural*: given any finite sample, it decomposes the regret into (a) the size of the set where the estimated rule disagrees with the oracle and (b) the oracle score mass on that set, identifying which component's estimation error is doing the damage. The two results are complementary, not competing.*

5.2 The Value-Conservatism Trade-off

The operator's preference weight g is a dial they can turn. Turning it away from the neutral setting ($g \equiv 1$, i.e., weight every individual equally regardless of risk) lets them exclude segments or concentrate treatment on others. The question is: what does turning that dial cost?

We measure cost in *unmodulated welfare*, written $\mathcal{W}(\pi; \mathbf{1})$: welfare evaluated as if the preference weight were neutral even when the operator chose a non-neutral one. This is the measure that tells us how much benefit the operator's chosen rule actually generates for the population,

ignoring how the operator feels about the distribution of that benefit. Proposition 3 gives a clean bound: any non-neutral preference weakly reduces unmodulated welfare. This is the central practical tool of the framework. It lets the operator make an informed decision about whether the targeting control they want is worth its welfare cost.

Definition 4 (Risk exposure)**.** *The risk exposure of a treatment rule π is the expected value of the risk covariate among treated individuals:*

$$\rho(\pi) = \mathbb{E}\big[R_i \mid \pi(X_i) = 1\big]. \tag{9}$$

Proposition 3 (Value-conservatism trade-off)**.** *Under Assumptions 1 and 2, let π_0^* denote the optimal rule under $g_0 \equiv 1$ (no risk modulation) and π_α^* the optimal rule under a risk-modulation function g_α with $0 \le g_\alpha(r) \le 1$ for all r. Then the unrestricted welfare satisfies*

$$\mathcal{W}(\pi_\alpha^*; \mathbf{1}) \le \mathcal{W}(\pi_0^*; \mathbf{1}). \tag{10}$$

That is, introducing any non-trivial risk modulation ($g_\alpha \not\equiv 1$) weakly reduces the unrestricted welfare relative to the unmodulated optimum. Equality holds if and only if g_α does not change the top-κ set: the set of individuals targeted by π_α^* is identical to the set targeted by π_0^*.

Proof. The rule π_0^* maximises $\mathcal{W}(\pi; \mathbf{1}) = \mathbb{E}[\pi(X_i) \cdot \tau(X_i) \cdot V_i]$ over Π_κ by construction. The rule π_α^* maximises a different objective, $\mathcal{W}(\pi; g_\alpha) = \mathbb{E}[\pi(X_i) \cdot \tau(X_i) \cdot V_i \cdot g_\alpha(R_i)]$. Since $\pi_\alpha^* \in \Pi_\kappa$ is feasible for the unrestricted problem but is not in general optimal for it, we have $\mathcal{W}(\pi_\alpha^*; \mathbf{1}) \le \mathcal{W}(\pi_0^*; \mathbf{1})$. For the equality condition: $\mathcal{W}(\pi_\alpha^*; \mathbf{1}) = \mathcal{W}(\pi_0^*; \mathbf{1})$ if and only if π_α^* is also optimal for the unrestricted welfare, which under Assumption 2 (unique threshold) holds if and only if the top-κ sets coincide. □

Remark 10 (Welfare-risk frontier)**.** *As the operator varies the preference function g, each choice produces a different pair (welfare, risk exposure). Plotting all such pairs traces out a curve in the welfare-risk plane, analogous to a Pareto frontier in multi-objective optimisation (a curve showing the best achievable value on one objective for each level of the other). Proposition 3 guarantees that the unmodulated rule ($g \equiv 1$) sits at the highest-welfare point on this curve. It does not guarantee that the curve is monotone in any particular dial the operator turns; a small change to g's shape can move both welfare and risk exposure in complicated ways.*

Remark 11 (When g can offset estimation error)**.** *Proposition 3 is a statement about oracle-optimal rules (rules that use the truth). Once we switch to using estimates, the story is not quite as clean. Suppose the true treatment effect factors multiplicatively as $\tau(x) = \tau_0(x) \cdot m(R(x))$, where τ_0 is a "base" effect that depends on covariates other than the risk variable, and $m : [0,1] \to [0,1]$ is a* moderation function *that says how the treatment effect is attenuated by the risk covariate (for instance, a treatment is most effective at moderate risk and vanishes at the extremes). Suppose further that the estimated treatment-effect model cannot capture m (because its feature set or flexibility is too limited). Then inserting $g = m$ into the composite score yields $\hat{\tau}(x) \cdot \hat{V}(x) \cdot m(\hat{R}(x))$, which approximates the true ranking to the extent that each estimated component tracks the rank contribution of its oracle counterpart. In this regime, the estimated rule that uses a non-neutral g may beat the estimated rule that uses a neutral g, because the operator's preference factor is silently compensating for what the treatment-effect model missed. This is a secondary use of the framework, not its primary justification: it only applies when g happens to mirror a real structural interaction the treatment-effect model missed, and the compensation is partial because m itself has to be approximated. The simulation in Section 7 does not exhibit this effect: even under the CATE misspecification of Section 7.3, calibration selects $g \approx 1$. The reason is that the misspecification tested there is feature dropout in the CATE model, which the value model partially absorbs through features it retains; this is outside the regime*

described above, which requires the operator to supply a g that mirrors a structural interaction no estimated component can recover. The Remark's regime is therefore a theoretical case in which g can help, not a claim about the DGP used in this paper, and the welfare improvement observed in Section 7.3 comes from value weighting, not from g.

6 Instantiation: Online Gaming

We now instantiate the general framework for retention bonus targeting in online gaming. The treatment is a binary retention bonus ($W_i = 1$: bonus received; $W_i = 0$: no bonus). The outcome is binary retention ($Y_i = 1$: player remains active). The retention window (the period over which "active" is measured, e.g., 7 days, 30 days, or 90 days) is an operator choice; all results hold for whichever window is used, but operators should note that treatment effects can differ across windows (a bonus that retains for a week may accelerate longer-term churn through habituation), and cross-window effects are a separate modelling concern beyond the scope of this framework. The outcome value is player lifetime value ($V_i = \text{LTV}_i$). The risk covariate is churn probability ($R_i = p_i^{\text{churn}}$). Throughout this section and Section 7 we use p in place of R for the risk covariate, to keep the notation close to the domain convention of referring to churn probability as p; the symbols denote the same quantity.

6.1 The Risk-Modulation Function

The effectiveness of a retention bonus depends on the player's position along the churn risk spectrum. Domain reasoning motivates three operational regimes:

1. **Low churn risk** ($p < p_L$). The player is stable and would likely remain active without intervention. A bonus consumes budget without generating incremental retention. The risk-modulation weight should be zero.

2. **Moderate churn risk** ($p \in [p_L, p_H]$). The player is at risk but reachable; a well-timed bonus can plausibly tip the retention decision. The weight should be at its maximum.

3. **High churn risk** ($p > p_H$). The player has likely disengaged to the point where a bonus cannot reverse the trajectory (Ascarza, 2018). The weight should be reduced, though a non-zero floor acknowledges uncertainty in the churn estimate.

We use a five-parameter piecewise linear g that encodes these regimes with smooth transitions. Because the specific functional form is an implementation choice rather than part of the contribution, we defer the formal definition and its regularity (Lipschitz) properties to Appendix A and keep only the operational reading here. The parameters $\theta = (p_L, p_{ML}, p_{MH}, p_H, \phi)$ have direct operational meanings: p_L is the churn probability below which players are excluded (budget conservation); p_{ML} and p_{MH} bound the "fully reachable" plateau where the weight is one; p_H is the churn probability above which the weight stops falling; and ϕ is the floor weight for the highest-risk players. Figure 1 plots g at the default values $\theta_0 = (0.05, 0.40, 0.60, 0.90, 0.20)$ used in the simulation. These defaults are pedagogical (chosen to produce visually clear contrasts across the regimes), not operational recommendations; real operators should set θ from domain evidence (for instance, p_L tends to be higher, 0.10 to 0.20, in real campaigns because "very stable player" is a commercial judgment, not a statistical one). The frontier analysis in Section 7.6 reports welfare costs for four alternative configurations (wide, default, narrow, aggressive); an operator can reproduce the same analysis for whichever θ is under consideration and read off the corresponding welfare cost.

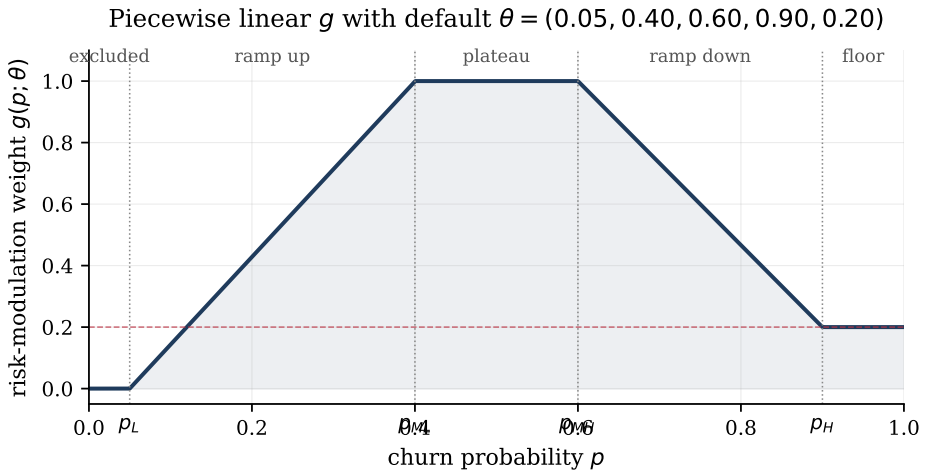


Figure 1: The piecewise linear risk-modulation function at default parameters $\theta_0 = (0.05, 0.40, 0.60, 0.90, 0.20)$. Players below p_L are excluded. Weight ramps to one through the moderate-risk plateau and falls to the floor ϕ for the highest-churn players. Formal definition in Appendix A.

6.2 Responsible-Gaming Considerations

The framework supports a pre-targeting *exclusion set* $\mathcal{E}$: a pre-specified list of individuals who are removed from the pool before ranking, regardless of their composite score. Writing $\mathcal{S}^*$ for the final set of individuals treated, the rule becomes "apply top-k ranking among individuals not in $\mathcal{E}$", or in notation $\mathcal{S}^* = \arg\text{top-k}_{i \notin \mathcal{E}}\, s_i$. In the gaming context, $\mathcal{E}$ should be determined by validated vulnerability screening instruments aligned with jurisdiction-specific regulatory requirements (e.g., the Problem Gambling Severity Index and self-exclusion registries in most jurisdictions; affordability checks and Gambling Commission licensing conditions in the United Kingdom; operator-side constraints from advertising regulators such as the Advertising Standards Authority or comparable bodies in other markets). This exclusion operates *above* the scoring framework: the welfare-optimal rule from Proposition 1 applies to the eligible population $\{1, \ldots, N\} \setminus \mathcal{E}$.

The floor parameter $\phi > 0$ means the framework assigns non-zero targeting weight to the highest-churn players. Some of these players may be churning because gambling is causing them harm; a production system should incorporate a "should we intervene?" gate informed by harm indicators beyond the scope of this framework, prior to the "who should we target?" ranking. The risk-modulation function encodes the operator's *targeting preferences*, not claims about *player welfare*. Setting $p_L > 0$ excludes stable players (budget conservation); reducing ϕ toward zero limits intervention on very high-churn players, some of whom may be disengaging for welfare-relevant reasons. The overlap is a side-effect of an operator-preference setting, not a principled harm gate: responsibility for harm-sensitive exclusions sits with $\mathcal{E}$ and the harm-indicator safeguards discussed below. However, the framework does not replace dedicated responsible-gaming safeguards: the exclusion set $\mathcal{E}$ requires domain expertise and regulatory alignment beyond the scope of this paper.

A related concern deserves its own flag. The V-weighting step, which is central to the paper's welfare-optimal result, multiplies the treatment-effect score by customer lifetime value. In online gaming this can compound harm: a player's lifetime value can be large precisely because they are gambling in harmful volumes, and value-weighted targeting will then concentrate bonuses on exactly those players. Nothing in the composite score detects this. We therefore treat two safeguards as preconditions, not options, for any responsible-gaming deployment of this

framework: (a) the exclusion set $\mathcal{E}$, populated from validated vulnerability signals, and (b) a harm gate that operates prior to (or in addition to) $\mathcal{E}$, using behavioural harm indicators (deposit velocity, session duration escalation, loss-chasing patterns) that are outside the scope of this framework. The composite-score ranking should not be deployed without both. Two further deployment defaults follow. First, the system should *fail closed*: if harm signals are unavailable for part of the population, the high-value tail of that part should be excluded by default rather than scored unguarded, so that missing harm data never silently reverts the rule to the unprotected score $\tau V g$. Second, membership of $\mathcal{E}$ and the harm signals feeding h must be re-evaluated at each campaign, not cached: vulnerability is dynamic, and a player outside $\mathcal{E}$ at one scoring round may have crossed a harm threshold by the next. Relatedly, although we use a non-zero floor $\phi = 0.20$ in the simulation for illustrative contrast, we recommend a default of $\phi = 0$ (no targeting weight on the highest-churn players) for live deployment, with any positive ϕ requiring explicit justification, since the highest-churn players are the most likely to be disengaging for welfare-relevant reasons; in some jurisdictions, direct promotion to identified at-risk players may also be restricted, which independently favours $\phi = 0$. Two ways to operationalise this within the existing machinery: (i) cap V_i above a threshold calibrated against harm indicators, so that the composite score stops growing with value once harm signals are present; or (ii) multiply V_i by a separate harm-protection factor that downweights individuals whose high value correlates with harm signals, logically distinct from $g(R_i)$ because it operates on the value dimension rather than the risk dimension. Both approaches admit a clean formal treatment as a fourth multiplicative factor in the welfare function, so harm protection can be an integral component of the ranking rather than an adjacent override.

Corollary 1 (Harm-protected welfare-optimal rule)**.** *Let $h : \mathcal{X} \to [0, 1]$ be a measurable* harm-protection weight *defined on the pre-treatment covariate vector, and define the extended welfare function*

$$\mathcal{W}(\pi; g, h) = \mathbb{E}\big[\pi(X_i) \cdot \tau(X_i) \cdot V_i \cdot g(R_i) \cdot h(X_i)\big]. \tag{11}$$

Under Assumptions 1 and 2 (with the regularity condition applied to the four-factor score), the treatment rule that maximises $\mathcal{W}(\pi; g, h)$ subject to $\pi \in \Pi_\kappa$ ranks individuals by the four-factor composite score $s^\dagger(x) = \tau(x) \cdot V(x) \cdot g(R(x)) \cdot h(x)$. Further, if $\pi^*_{g,h}$ denotes the welfare-optimal rule for given (g, h) and $\pi^*_{1,1}$ the rule under $g \equiv 1, h \equiv 1$, then

$$\mathcal{W}(\pi^*_{g,h}; \mathbf{1}, \mathbf{1}) \;\le\; \mathcal{W}(\pi^*_{1,1}; \mathbf{1}, \mathbf{1}), \tag{12}$$

with equality if and only if the top-κ sets coincide.

Proof. Both claims follow from the proofs of Propositions 1 and 3 applied to the four-factor integrand: the Lagrangian argument pins π^* down pointwise on $s^\dagger(x)$, and the feasibility argument for (12) mirrors that of Proposition 3 with $g \cdot h$ in place of g_α. □

Value capping is the special case of Corollary 1 in which $h(x)$ equals $\bar{V}/V(x)$ on the capped set and 1 elsewhere, so the product $V(x) \cdot h(x)$ equals $\min(V(x), \bar{V})$; the multiplicative harm downweight is the general case. Equation (12) quantifies the welfare cost of harm protection on the same yardstick used for g, so an operator can read it off the cost-quantification procedure of Section 7 without new machinery.

A soft weight does not guarantee protection; a hard constraint does. The multiplicative weight h is *soft*: a sufficiently large $\tau(x)V(x)$ can still clear the threshold despite a small $h(x)$, so h cannot guarantee any ceiling on the harm exposure of the treated set. When the operator needs a guarantee, the right object is a constraint, not a weight. Add to the budget a harm-exposure cap and solve

$$\max_{\pi \in \Pi_\kappa} \mathbb{E}\big[\pi(X_i)\, \tau(X_i) V_i g(R_i)\big] \quad \text{s.t.} \quad \mathbb{E}\big[\pi(X_i)\, \mathrm{harm}(X_i)\big] \le \bar{H}\, \kappa, \tag{13}$$

where harm(x) is a harm indicator and $\bar{H}$ caps the mean harm among the treated.

Corollary 2 (Harm-constrained welfare-optimal rule)**.** *Under Assumptions 1 and 2, the rule solving* (13) *ranks individuals by the harm-penalised score*

$$s^{\ddagger}(x) = \tau(x)V(x)g(R(x)) - \mu\,\mathrm{harm}(x), \tag{14}$$

treating those with $s^{\ddagger}(x) > \lambda$, where $\lambda \geq 0$ is the budget multiplier and $\mu \geq 0$ is the smallest value for which the harm constraint holds (with $\mu = 0$ when the constraint slacks). For any target harm level $\bar{H}$ achievable within the budget, this rule attains weakly higher welfare than any *rule meeting the same harm level, including a soft-weight rule h tuned to that level.*

Proof. Attach multipliers λ (budget) and $\mu \geq 0$ (harm) to the two constraints; the Lagrangian integrand is $\pi(x)\,[\tau V g - \mu\,\mathrm{harm} - \lambda]$, maximised pointwise by treating exactly when $s^{\ddagger}(x) = \tau V g - \mu\,\mathrm{harm} > \lambda$ (the same Neyman-Pearson argument as Proposition 1, now on the penalised score). Strong duality holds for the convex relaxation by the same Slater argument (the two-constraint feasible region is non-empty in its interior whenever a strictly feasible rule exists), and the relaxed optimum is again attained by a binary rule almost surely, because the boundary set $\{s^{\ddagger}(x) = \lambda\}$ is measure-zero under Assumption 2(i) applied to the penalised score. The dominance claim is immediate: the rule solves (13), so it is the welfare-maximiser over all feasible rules meeting the harm cap; a soft-weight rule achieving the same harm level is one such feasible rule, hence weakly worse. □

The practical reading: reserve the soft weight h for graded, sub-threshold harm signals where some trade-off against value is acceptable, and use the hard constraint (13) (or membership of the exclusion set $\mathcal{E}$, the $\bar{H} = 0$ limit) when a guarantee is required.

A worked harm-gate result. To make the trade-off concrete we add a harm covariate to the simulation, drawn so that harm rises with spend and therefore correlates positively with lifetime value (the configuration the paper warns about: high value can co-occur with high harm). Population mean harm is 0.42. We compare the unguarded value-weighted rule against the harm-protected rule that multiplies by a weight h falling from 1 to 0 as harm rises from 0.6 to 0.9, at $\kappa = 0.30$ over 200 replications. For the oracle rule (true components), harm protection reduces mean harm exposure among the treated from 0.466 to 0.411 (an 11.8% reduction) at a welfare cost of 6.4% (44,615 → 41,776): this is Corollary 1 made concrete, with the welfare cost read off the same yardstick as g. Confirming the warning at the head of this subsection, the unguarded score concentrates harm *above* the population mean (0.466 vs. 0.42), so without the gate value weighting does pull treatment toward higher-harm players. For the estimated rule the harm-exposure reduction is similar (0.475 → 0.411, about 13%) but the welfare *cost* is negative (16,824 → 19,400): in this DGP harm protection happens to remove high-value, low-effect players that the estimated score had over-ranked, so it improves operator welfare while reducing harm. That sign flip is the estimation-regime analogue of Remark 11 and should not be read as a general guarantee that harm protection is free; the oracle result is the reliable statement of its cost. Corollary 2 sharpens the design choice. Holding the harm-exposure level fixed at the 0.411 the soft gate achieves, the hard harm constraint of (13) attains welfare 43,552 against the soft gate's 41,776 (oracle, 200 replications), cutting the welfare cost of reaching that harm level from 2,839 to 1,063, a 62% reduction. The hard constraint sits on the welfare-harm efficient frontier that the soft multiplicative weight does not: when a guaranteed harm ceiling is the goal, the constraint is both safer (it cannot be overridden by a large τV) and cheaper in welfare than the soft weight.

Welfare-optimal targeting in the econometric sense is not the same as ethically optimal targeting in a harm-sensitive domain, and operators carry the responsibility for that distinction. We adopt

the phrase *operator welfare* for the econometric quantity maximised by the composite score and reserve *player welfare* for the wellbeing of the individuals targeted; this paper's propositions are about operator welfare, while responsible-gaming safeguards protect player welfare, and the two can point in different directions.

6.3 Numerical Example

Example 1 (Illustrative targeting)**.** *Consider eight hypothetical players with known treatment effects, lifetime values, and churn risk scores (Table 2). The values in Table 2 are illustrative, chosen so that the three scoring rules produce visibly contrasting top-k sets. They do not attempt to respect the tight bound $\tau \leq p$ that a strictly binary-retention reading would impose (τ for a player with churn probability* 0.03 *is bounded above by* 0.03*). Readers should treat the numbers as abstract scores sufficient to illustrate ranking behaviour, not as a realistic simulation. We apply the composite score $s_i = \tau_i \cdot V_i \cdot g(p_i; \theta_0)$ with default parameters $\theta_0 = (0.05, 0.40, 0.60, 0.90, 0.20)$ and a budget of* 3 *out of* 8 *players.*

Table 2: Numerical example: eight hypothetical players. Columns show the treatment effect (τ, in percentage points; values are illustrative and do not attempt to respect the bound $\tau \leq p$ that a strictly binary-retention reading would impose, see text above), lifetime value (V), churn risk score (p), the risk-modulation weight $g(p)$, the composite score s, and the targeting rank under three scoring rules. Budget: 3 players. The composite score s here multiplies τ in percentage points for readability, so the magnitudes are a factor of 100 larger than they would be if τ entered as a probability; ranks are unaffected. The welfare values reported in Section 7 use τ as a probability.

Player	τ (pp)	V	p	$g(p)$	s	Rank: τ	Rank: τV	Rank: $\tau V g$
A	+35	200	0.03	0.00	0	1	4	–
B	+30	500	0.25	0.57	8,571	2	2	3
C	+20	800	0.50	1.00	16,000	3	1	1
D	+15	600	0.45	1.00	9,000	4	3	2
E	+10	400	0.80	0.47	1,867	5	6	4
F	+5	1,200	0.10	0.14	857	7	5	5
G	−10	900	0.50	1.00	−9,000	8	8	8
H	+8	300	0.95	0.20	480	6	7	6

The example illustrates three key effects:

Pure uplift (τ alone) *selects players A, B, C. Player A has the highest treatment effect but very low churn risk ($p = 0.03$): they would retain without a bonus. The budget is wasted on a "sure thing."*

Value-weighted uplift ($\tau \times V$) *selects C, B, D. Player A drops from first to fourth because their LTV is low (200 vs. 500–800 for B and C). This rule correctly prioritises high-value responders but treats all churn risk levels equally.*

Full composite ($\tau \times V \times g$) *selects C, D, B. Player A is vetoed entirely ($g = 0$: churn too low to justify intervention). Player E is deprioritised ($g = 0.47$: high churn reduces the weight). Players C and D, both in the moderate-churn plateau ($g = 1$), dominate. The operator concentrates budget on the segment where interventions are judged most effective.*

Player G is a "sleeping dog": an individual made worse off by treatment ($\tau < 0$). All three rules correctly give this player a negative score and exclude them from targeting. None of the three rules considers harm: a harm-sensitive operator would, in addition, apply the harm-protection weight h of Section 6.2 (and the exclusion set $\mathcal{E}$) before ranking, which can demote an otherwise high-scoring high-value player; the worked harm-gate result in Section 6.2 quantifies that effect.

7 Simulation Study

We evaluate the framework in a controlled simulation. The advantage of a simulation over a real data set is that we know the true treatment effects, true outcome values, and true risk-covariate values for every individual, so we can compare what each targeting rule achieves to the best-possible-given-the-truth benchmark. The trade-off is that the simulation's validity rests on our simulation world resembling a real targeting problem closely enough to be informative. We return to this point in Section 8.

The simulation is organised around three scenarios, each isolating when one component of the composite score adds value: (i) a well-specified *data-generating process* (DGP, the set of equations we use to generate the simulated individuals and outcomes) in which the treatment-effect estimator has access to all relevant features; (ii) a misspecified DGP in which the key feature driving the treatment-effect heterogeneity is hidden from the treatment-effect model; and (iii) a null DGP in which the treatment effect does not interact with the risk covariate at all. Section 7.2 presents the well-specified scenario. Sections 7.3 and 7.4 cover treatment-effect misspecification and the null DGP. Sections 7.5–7.7 report supporting analyses: budget sensitivity, the welfare-risk frontier, and robustness checks.

7.1 Design

We generate a simulated population of $N = 10{,}000$ individuals, each with four covariates. Treatment effects are nonlinear functions of the covariates, and the covariates also drive outcome value and churn probability, so the three components are correlated through the covariates as they are in practice. The full data-generating process follows.

Features. We generate four pre-treatment covariates independently of one another, one per individual:

$$\begin{aligned}
\text{account_age}_i &\sim \mathcal{N}(24, 8), \quad \text{clipped to } [1, 60], \\
\text{weekly_spend}_i &\sim \exp(\mathcal{N}(3.5, 0.8)), \quad \text{clipped to } [5, 500], \\
\text{login_freq}_i &\sim \text{Poisson}(4), \quad \text{clipped to } [1, 15], \\
\text{bonus_history}_i &\sim \mathcal{N}(3, 1.5), \quad \text{clipped to } [0, 10],
\end{aligned}$$

where $\mathcal{N}(\mu, \sigma)$ is the normal (Gaussian) distribution with mean μ and standard deviation σ (a bell-shaped distribution centred at μ with spread σ), $\exp(\mathcal{N}(\cdot, \cdot))$ is the log-normal (exponentiating a normal gives a right-skewed distribution with no negative values, appropriate for strictly positive quantities like spend), and Poisson(λ) is the Poisson distribution with mean λ (used for non-negative integer counts like login frequency). "Clipped to $[a, b]$" means any value falling outside the range $[a, b]$ is pulled in to the nearest boundary, so the variable is guaranteed to live inside realistic bounds. This creates a small amount of probability mass at each endpoint; downstream estimators can in principle pick up on this concentration, though the effect is small at the clipping thresholds used here.

Let z_{spend}, z_{login}, z_{age}, z_{bonus} denote the *standardised* versions of each feature. Standardisation is the operation of subtracting the feature's mean and dividing by its standard deviation, so that the resulting variable has mean 0 and standard deviation 1; this makes the coefficient on each feature in the equations below directly comparable across features. We compute the means and standard deviations once from a very large sample ($N = 2{,}000{,}000$ draws) rather than from each simulation replication's own data. This is a *held-out standardisation*: it prevents the training sample's noise from leaking into the test sample's coding of the features, a subtle source of bias when the standardisation is re-computed on each training set.

Churn probability. We generate each individual's churn probability from a *logistic model*, a standard way of producing a value between 0 and 1 as a function of covariates. The model

first forms a linear combination of the covariates (called the *log-odds* or *logit*, because it equals $\log(p/(1-p))$ for a probability p):

$$\text{logit}(p_i) = -1.0 + 0.04\,(30 - \text{age}_i) - 0.25\,\text{login}_i + 0.01\,(40 - \text{spend}_i) + 0.1\,z_{\text{bonus},i} + \varepsilon_i^{(p)}, \quad (15)$$

where $\varepsilon_i^{(p)} \sim \mathcal{N}(0, 0.4)$ is an individual-level noise term. The scaling is deliberately mixed: age_i and spend_i enter as offsets from constants (30 and 40) so that the intercept reflects churn at reference values for those features rather than at zero; login_i enters in raw units, appropriate for a count with a natural scale; and bonus_history_i enters standardised so that its coefficient is unit-free. The offsets (30, 40) are pedagogical anchors, not the feature means; they shift the intercept but leave the covariance structure unchanged. We then convert the log-odds to a probability with the *sigmoid* (or inverse-logit) function $\text{sigmoid}(z) = 1/(1+e^{-z})$, which is a smooth S-shaped curve mapping any real number to the $(0, 1)$ interval: $p_i = \text{sigmoid}(\text{logit}(p_i))$. We then clip p_i to $[0.01, 0.99]$ to avoid degenerate extremes.

Lifetime value.

$$V_i = 10\,\text{spend}_i + 40\,\text{login}_i + 5\,\text{age}_i + \varepsilon_i^{(V)}, \quad \varepsilon_i^{(V)} \sim \mathcal{N}(0, 80), \quad \text{clipped to } [50, 8000]. \quad (16)$$

CATE with churn moderation. The treatment effect features an inverse-U interaction with churn probability:

$$\begin{aligned} \tau_{\text{base},i} &= 0.06 + 0.10\,z_{\text{login},i} - 0.02\,z_{\text{login},i}^2 - 0.04\,z_{\text{spend},i}, \\ m(p_i) &= 4\,p_i\,(1-p_i), \\ \tau_i &= \tau_{\text{base},i} \cdot m(p_i) + \varepsilon_i^{(\tau)}, \quad \varepsilon_i^{(\tau)} \sim \mathcal{N}(0, 0.03), \quad \text{clipped to } [-0.20, 0.40]. \end{aligned} \quad (17)$$

The moderation function $m(p) = 4p(1-p)$ peaks at $p = 0.5$ and equals zero at $p \in \{0, 1\}$, creating the structural feature that risk modulation is designed to exploit: treatment effects are highest at moderate churn and attenuated at both extremes. The symmetric shape is a simulation convenience: Ascarza (2018) documents asymmetric attenuation in real retention settings (treatment effects tend to fall off more sharply at high churn than at low churn), and an operator using this framework on real data should not assume the moderation is symmetric.

Baseline and outcomes. Each individual has a baseline retention probability (the probability they would remain active if not treated) drawn from a *Bernoulli* distribution (a single coin flip with a specified probability of success):

$$Y_i(0) \sim \text{Bernoulli}\big((0.6 + 0.15\,z_{\text{login},i} + 0.05\,z_{\text{spend},i} - 0.08\,z_{\text{age},i} + \varepsilon_i^{(b)})_{[0.05,\,0.95]}\big), \quad (18)$$

with $\varepsilon_i^{(b)} \sim \mathcal{N}(0, 0.05)$. Let baseline_i denote the clipped linear predictor (the expression inside the Bernoulli). The potential outcome under treatment is $Y_i(1) \sim \text{Bernoulli}((\text{baseline}_i + \tau_i)_{[0.01,\,0.99]})$: treatment simply shifts the retention probability by the treatment effect τ_i. For individuals with baseline retention close to 1 and positive τ_i, the clip to $[0.01, 0.99]$ can activate, so the *effective* treatment-effect realised in the outcome is smaller than the nominal τ_i. Welfare on the test set is evaluated using nominal $\tau_i \cdot V_i$ (the DGP's generated treatment effect), not the clip-adjusted effective value, so reported welfare numbers overstate the realised lift for high-baseline individuals. This bias is not uniform across strategies: because high baseline retention correlates with high login, spend, and hence high V (Equation (16)), any strategy that concentrates on the high-V tail is flattered most. This includes value-weighted and joint-model strategies, so the bias points, in principle, toward inflating the very comparisons the paper foregrounds. A clip-adjusted (effective-τ) re-evaluation of the headline table, reported in Section 7.7, confirms the bias is modest (3.7% to 6.7% across strategies) and, although mildly larger for the value-weighted and joint-model strategies as anticipated, does not change any ranking. We assign

treatment with a fair coin flip, $W_i \sim \text{Bernoulli}(0.5)$, mimicking a randomised experiment. This design trivially satisfies the overlap condition of Assumption 1(ii) with $\eta = 0.5$ and does not stress-test the low-propensity regime typical of observational deployments, where $P(W = 1 \mid X)$ can sit well below 0.5; robustness under non-trivial propensity variation is outside the scope of the present simulations.

Estimation protocol. We run 200 independent replications of the following protocol to characterise variability across simulation runs. Each replication $r \in \{0, \ldots, 199\}$ uses the fixed seed $42 + 7r$, so the entire study is deterministic and reproducible; the simulation code, the revision-additions code (asymmetric DGP, harm gate, effective-welfare, and the S-Learner under misspecification and the null DGP), and the resulting output files are released in the companion repository.

Train/test split. We partition the population 70/30 into a training set (used to fit all models) and a test set (used only for final evaluation). This prevents each strategy from inadvertently profiting from noise in the data it was trained on, a practice known in ML as out-of-sample evaluation.

Component models. We estimate three models on the training data: a *T-Learner* for the treatment effect, which fits one outcome model on the treated group and one on the control group using gradient boosted trees (an ensemble ML method that combines many shallow decision trees, each correcting the errors of the previous ones) and takes the difference; a gradient boosted regression model for outcome value; and a gradient boosted classifier for churn probability (a similar ensemble, predicting a probability instead of a continuous value).

Value-model misspecification. We deliberately misspecify the value model: the training target uses the form $12 \cdot \text{spend}^{0.9} + 35 \cdot \text{login} + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, 200)$, which drops the age feature and uses a nonlinear transformation of spend. This mimics the realistic situation where the true value function is unknown and the analyst has to pick a plausible but wrong functional form.

Churn model. The churn model is trained on the control-group non-retention indicator, $1 - Y_i \mid W_i = 0$ (one minus the retention outcome for individuals who did not receive the bonus). This is a reasonable proxy for churn risk but is not identical to the structural churn probability used in the DGP; see Remark 12.

Evaluation. All component models are estimated on training data alone. The true ("oracle") values of τ, V, and R, known to us because we generated them, are used only to compute welfare on the test set and to serve as a theoretical upper bound on what any estimated rule could achieve. We report two reference budgets rather than one. The realistic operating budget is $\kappa = 0.10$ (treat the top 10% of the test population), which matches real marketing-operations practice, where a single retention campaign typically reaches 5% to 15% of the active base; this is the budget on which an operator's deployment decision actually turns. We also report $\kappa = 0.30$ as a diagnostic budget: it gives every strategy enough latitude to express its preferences over the population, which makes the mechanism behind each result easier to see, but it is more generous than most campaigns. The two budgets tell different stories, and the difference is itself a finding (Section 7.5): Section 7.5 varies κ across $\{0.10, 0.20, 0.30, 0.50\}$, and the ranking of strategies depends on κ.

Robustness. As an additional check we re-estimate the treatment effect using an *X-Learner* (Künzel et al., 2019), a metalearner that first imputes individual-level treatment-effect estimates by combining each unit's observed outcome with a model-predicted counterfactual, then fits separate treatment-effect models on the treated and control groups; results appear in Section 7.7.

Calibrated parameter selection. The preference-weight parameters θ are picked by held-out grid search. A *grid search* is an exhaustive scan over a fixed list of candidate parameter

settings; *held-out* means we evaluate each candidate on a slice of data that was not used to fit the models it relies on, which keeps the selection honest. Concretely, on each replication we further split the training set 60/40 into an *estimation fold* (used to fit the three component models) and a *validation fold* (used only to score the candidate θ values). We test 22 parameter configurations, including a near-identity configuration with $p_L = 0$, $p_{ML} = 0.01$, $p_{MH} = 0.99$, $p_H = 1$, $\phi = 1$. Under this parameterisation $g(p) = 1$ everywhere except on a vanishingly small ramp $p \in [0, 0.01)$, so it effectively turns g into the constant function 1 and lets calibration decline any preference weighting if it does not help. For each candidate, we compute the composite score on the validation fold using the estimated components $(\hat{\tau}, \hat{V}, \hat{R})$, apply the resulting rule, and measure its welfare. We pick the configuration with the highest validation-fold welfare. No oracle information enters parameter selection.

Statistical testing. We compare strategies using *paired t-tests* across the 200 replications: on each replication each strategy produces a welfare number, and the t-test asks whether one strategy's mean welfare reliably exceeds another's across replications ("paired" because we compare strategies on the same replications, which removes replication-to-replication variation from the noise). Each scenario (well-specified, misspecified, null DGP, budget sensitivity, robustness) defines its own family of comparisons. Within each family we apply the *Holm-Bonferroni correction*: a sequential procedure that sorts the raw p-values in ascending order and inflates each one by a progressively smaller factor, controlling the probability of even one false positive across the whole family. We do not apply an additional correction across families, because each scenario targets a distinct empirical question (we treat each family's conclusions as independent). We also report *Cohen's d*, the standardised effect size (difference in means divided by pooled standard deviation), as a unit-free measure of how large each difference is. Because the paired structure removes replication-to-replication variation from the noise, even small mean differences can be highly significant, so we report *Cohen's d* alongside every p-value and lean on it as the unit-free measure of magnitude. The relevant precision yardstick for a paired test is the standard deviation of the paired *differences*, not of the welfare levels. For the headline comparisons that standard deviation is on the order of 850 to 900 welfare units (for example, 891 for calibrated-vs-CATE-Only at $\kappa = 0.30$), so with 200 replications the minimum welfare difference detectable at $p < 0.05$ is roughly 120 to 125 units, about 0.7% of the $\kappa = 0.30$ value-weighted welfare. Differences of this order are therefore statistically resolvable but operationally small; we report them with their effect sizes and do not treat a small but resolvable gap as a large one. The paired t-test assumes approximate normality of the per-replication differences; with 200 replications this is justified by the central limit theorem, and a Wilcoxon signed-rank check (not reported) leaves every reported conclusion unchanged.

Strategies evaluated. We compare seven targeting strategies: (i) *Random*: uniform random selection, providing a lower benchmark; (ii) *CATE Only*: rank by the estimated treatment effect $\hat{\tau}(x)$ alone; (iii) *CATE × LTV*: rank by the product of estimated treatment effect and estimated outcome value, $\hat{\tau}(x) \cdot \hat{V}(x)$, the current practitioner heuristic; (iv) *Full Framework (Est.)*: rank by $\hat{\tau}(x) \cdot \hat{V}(x) \cdot g(\hat{R}(x); \theta_0)$ with default (uncalibrated) preference parameters $\theta_0 = (0.05, 0.40, 0.60, 0.90, 0.20)$; (v) *Full Framework (Cal.)*: same as (iv) but with the preference parameters chosen by the held-out calibration procedure; (vi) *Full Framework (Oracle)*: the estimation-free benchmark for the decomposed framework at θ_0, ranking by the true (not estimated) components $\tau(x) \cdot V(x) \cdot g(R(x); \theta_0)$; because $g(\cdot; \theta_0) \not\equiv 1$, this is not the welfare-maximising rule under $\mathcal{W}(\pi; \mathbf{1})$ (by Proposition 3), but it bounds how much of the framework's welfare gap to the estimated versions is attributable to estimation error in $\hat{\tau}, \hat{V}, \hat{R}$; (vii) *S-Learner*: a single gradient boosted outcome model that takes treatment as an input feature (max tree depth 4, compared with 3 for the T-Learner), producing a treatment-effect estimate by subtracting its prediction at $W = 0$ from its prediction at $W = 1$: $\hat{\tau}_{\mathrm{SL}}(x) = \hat{f}(x, 1) - \hat{f}(x, 0)$, then ranking by $\hat{\tau}_{\mathrm{SL}}(x) \cdot \hat{V}(x)$ (Künzel et al., 2019). The S-Learner is a single-model policy

alternative; it does not use the three-component decomposition for the treatment-effect estimate, but it still borrows $\hat{V}$ from the decomposed pipeline (a fully end-to-end alternative would predict the value-weighted outcome directly). The deeper trees give the S-Learner more room to capture treatment-covariate interactions. Section 7.7 reports a depth-matched ablation (S-Learner at depth 3; T-Learner at depth 4 paired with the existing LTV model) which shows the joint-modelling advantage persists, and modestly widens, once the tree-depth hyperparameter is equalised in either direction.

7.2 Main Results

Table 3 reports the seven strategies side by side; Figure 2 plots the welfare column visually. All welfare figures use the oracle welfare function $\mathcal{W}(\pi; \mathbf{1}) = \sum_{i \in \text{targeted}} \tau_i \cdot V_i$ on the test set, giving every strategy a common yardstick. This choice has a consequence worth flagging: by Proposition 3, any strategy that optimises a surrogate with $g \neq 1$ weakly underperforms the $g \equiv 1$-optimal strategy under this yardstick. That is by design; an operator who chooses $g \neq 1$ is expressing a targeting preference and accepting the welfare cost the proposition quantifies.

Table 3: Targeting strategy comparison (N = 10,000, 200 replications, $\kappa = 0.30$). ATE is the average treatment effect among treated (percentage points). Welfare throughout is *operator* welfare, total value generated under $\mathcal{W}(\pi; \mathbf{1}) = \sum \tau_i \cdot V_i$ on the test set (every "welfare" figure in the paper is operator welfare in the sense of Section 6.2, not player wellbeing). Risk exposure is mean churn probability among treated. Churn Mod is the mean churn moderation $4p(1-p)$ among treated. Values shown as ± are standard deviations across replications; pairwise significance tests are in Table 4.

Strategy	ATE (pp)	Welfare	Risk Exp.	Churn Mod
Random	1.2 ± 2.1	$5{,}399 \pm 1{,}359$	0.166	0.512
CATE Only	3.3 ± 1.9	$16{,}634 \pm 2{,}954$	0.153	0.485
CATE × LTV	3.0 ± 2.0	$16{,}427 \pm 3{,}090$	0.136	0.440
Full Framework (Est.)	2.8 ± 2.0	$13{,}629 \pm 3{,}008$	0.146	0.468
Full Framework (Cal.)	3.0 ± 2.0	$16{,}427 \pm 3{,}090$	0.136	0.440
Full Framework (Oracle)	6.7 ± 2.0	$36{,}641 \pm 704$	0.192	0.596
τV Oracle ($g \equiv 1$)	6.8 ± 1.8	$44{,}615 \pm 805$	0.138	0.451
S-Learner	3.9 ± 2.0	$24{,}310 \pm 4{,}098$	0.120	0.399

Table 4: Pairwise welfare differences (Holm-Bonferroni corrected, 200 replications). Positive values favour the first strategy.

Comparison	Diff.	t	p_{holm}	Note
CATE Only vs. Random	+11,235	50.61	< 0.0001	–
CATE × LTV vs. Random	+11,029	48.38	< 0.0001	–
Cal. vs. CATE × LTV	−1	−0.97	0.333	calibration picks $g \approx 1$
Cal. vs. CATE Only	−207	−3.27	0.0026	$d = -0.231$
Default Est. vs. CATE × LTV	−2,799	−45.59	< 0.0001	default g costs welfare
Oracle vs. CATE × LTV	+20,214	92.74	< 0.0001	estimation is the bottleneck

Four patterns stand out. First, CATE-based targeting triples welfare relative to random allocation (16,634 vs. 5,399), confirming that heterogeneity in treatment effects is worth exploiting. Second, the calibration procedure selects the near-identity configuration in 197 of 200 replications, so the calibrated strategy reproduces CATE × LTV (16,427) almost exactly. This is the behaviour that Proposition 3 predicts for the oracle rule, and the calibration procedure tracks it empirically because estimation is accurate enough in this DGP: the calibration objective is estimated welfare under $g \equiv 1$, which Proposition 3 identifies as the welfare-maximising choice,

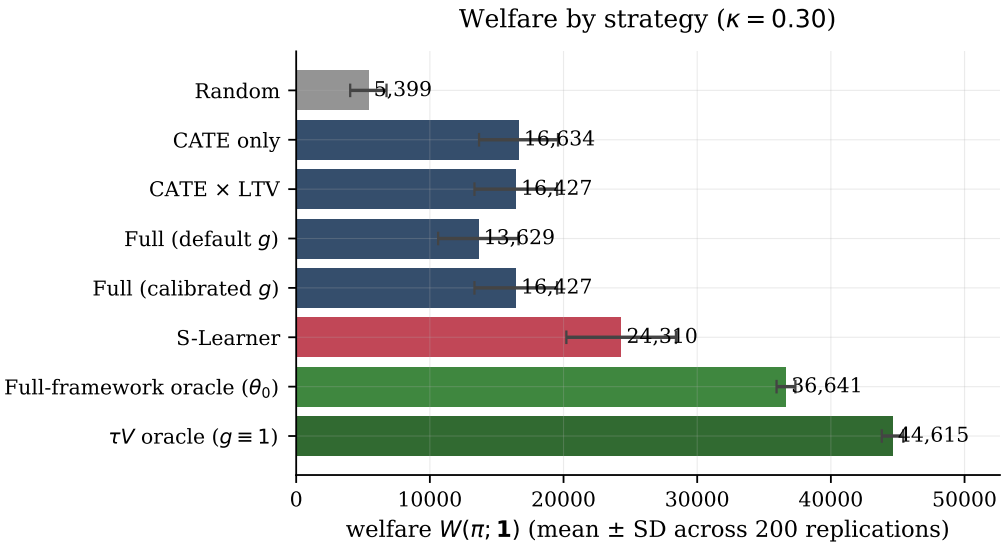


Figure 2: Welfare by targeting strategy ($\kappa = 0.30$, 200 replications). Error bars are ±1 standard deviation. The S-Learner (red) beats every decomposed estimated strategy; the Full-framework oracle (light green) is the estimation-free benchmark for the decomposed framework at θ_0; the τV oracle (dark green) is the welfare-maximising rule under $\mathcal{W}(\pi; \mathbf{1})$ ($g \equiv 1$) and the highest-welfare point achievable in this DGP.

so calibration approaches the identity as estimates approach truth. The calibration procedure's substantive output is the frontier of welfare costs across configurations (Section 7.6), which is a tool for pricing operator preferences, not for discovering them. Third, the default (uncalibrated) g costs 17% of welfare (13,629 vs. 16,427), making the value-conservatism trade-off quantitative at one representative parameterisation. By Proposition 3, any fixed-parameter g other than the identity imposes some welfare cost; 17% is simply the cost at the pedagogical θ_0, and a different θ would produce a different cost that the frontier analysis in Section 7.6 maps out. Note that the calibrated framework slightly underperforms pure CATE targeting at this budget: 16,427 vs. 16,634, a gap of 207 welfare units ($d = -0.231$, $p_{\text{holm}} = 0.0026$). This gap is statistically resolvable (it exceeds the $\sim$125-unit detection floor derived above, consistent with $p_{\text{holm}} = 0.0026$) but operationally small: 207 units is about 1.2% of welfare, and the effect size is small ($d = -0.231$). The magnitude matters less than the fact that its *sign* reverses with budget: at tighter budgets CATE Only underperforms CATE × LTV, and at looser budgets it overperforms (Section 7.5). That reversal, not the size of any single gap, is the substantive point. Fourth, the S-Learner achieves 24,310, substantially above every decomposed estimated strategy. The Full-framework oracle at θ_0 scores 36,641, and the τV oracle (the welfare-maximal rule under $\mathcal{W}(\pi; \mathbf{1})$, i.e. $g \equiv 1$) scores 44,615. The gap between these two (7,974 welfare units) is the cost of the default θ_0 preference alone, independent of any estimation error; the gap between the S-Learner and the τV oracle (20,305 units) is the estimation-error cost under the common $\mathcal{W}(\pi; \mathbf{1})$ yardstick. The S-Learner gap is the welfare-interpretability trade-off; we return to it in Section 8.

7.3 CATE Misspecification

Design. The CATE model is re-estimated with login frequency dropped from its feature set; LTV and churn models retain all four features. This simulates a realistic scenario where the uplift model is imperfect while auxiliary models are better specified. All other details are identical to the primary simulation.

Table 5: CATE misspecification results: the CATE model drops login frequency; LTV and churn models retain all features. Same DGP, $N = 10{,}000$, 200 replications, $\kappa = 0.30$. p-values are Holm-Bonferroni corrected. Risk exposure and churn moderation columns are omitted as they are qualitatively similar to the well-specified scenario.

Strategy	ATE (pp)	Welfare
CATE Only	1.1	$6{,}700 \pm 2{,}056$
CATE × LTV	1.4	$6{,}955 \pm 2{,}337$
Full Framework (Est.)	0.9	$3{,}809 \pm 2{,}339$
Full Framework (Cal.)	1.4	$6{,}954 \pm 2{,}336$
S-Learner	3.9	$24{,}310 \pm 4{,}098$
Cal. vs CATE Only: $+254$, $p < 0.0001$, $d = 0.303$ *(Cal. significantly better)*		
Cal. vs CATE × LTV: -1, $p = 0.42$, $d = -0.057$ *(Cal. = CATE × LTV,* $g \approx 1$ *selected)*		

All strategies degrade. CATE Only drops from 16,634 to 6,700 (a 60% welfare loss), confirming login frequency as the primary driver. The calibrated framework (6,954) outperforms CATE Only ($+254$, $p < 0.0001$, $d = 0.303$), and it matches CATE × LTV (6,954 vs. 6,955; not significant) because calibration again picks $g \approx 1$. The $+254$ gap is statistically resolvable (well above the ~ 125-unit detection floor of Section 7.1, with $d = 0.303$), though still operationally modest at roughly 3.8% of the misspecified-scenario welfare; we read it as a real but small improvement. So the improvement over CATE Only comes from value weighting, not from g. When the CATE model is degraded, the LTV model's features give $\hat{\tau} \cdot \hat{V}$ information the CATE estimator alone cannot reach; this is the mechanism described in Remark 11, operating in the limit where LTV alone absorbs the missing moderation structure. The S-Learner is unaffected by this misspecification: its welfare (24,310) is identical to the well-specified scenario, because the login-frequency dropout cripples only the decomposed T-Learner CATE, whereas the joint model retains all four features and never uses the decomposed estimate. This is a point against, not for, the decomposed framework's claimed "robustness under component misspecification": the joint model is the more robust of the two here, degrading not at all where the decomposition loses 60%.

7.4 Null DGP: No CATE-Risk Interaction

We repeat the simulation with $m(p) \equiv 1$ (no CATE-churn interaction), so that $\tau_i = \tau_{\text{base},i} + \varepsilon_i^{(\tau)}$. All other aspects remain identical.

Table 6: Null DGP results: no CATE-churn interaction ($m(p) \equiv 1$). Same estimation protocol, $N = 10{,}000$, 200 replications, $\kappa = 0.30$.

Strategy	Welfare
CATE Only	$63{,}098 \pm 5{,}549$
CATE × LTV	$68{,}256 \pm 6{,}324$
Full Framework (Est.)	$53{,}887 \pm 6{,}484$
Full Framework (Cal.)	$68{,}256 \pm 6{,}324$
S-Learner	$86{,}347 \pm 5{,}352$
Cal. vs CATE × LTV: diff $= 0$, *Cal. = CATE × LTV (*$g \approx 1$ *selected)*	

Under the null DGP, CATE × LTV (68,256) outperforms CATE Only (63,098), and the calibrated framework reproduces CATE × LTV because the calibration procedure selects the near-identity configuration as it must (again by Proposition 3). The substantive empirical content of this scenario is that value weighting beats pure CATE targeting even when there is no risk-moderation structure in the DGP to exploit. The S-Learner again leads (86,347), consistent

with its dominance in the main and misspecified scenarios.

7.5 Budget Sensitivity

Table 7 reports welfare across four budget levels.

Table 7: Budget sensitivity: welfare across budget fractions ($N = 10{,}000$, 200 replications).

Strategy	$\kappa = 0.10$	$\kappa = 0.20$	$\kappa = 0.30$	$\kappa = 0.50$
CATE Only	4,766	10,750	16,634	26,740
CATE × LTV	3,268	9,885	16,427	26,941
Full Framework (Est.)	2,586	7,960	13,629	25,171
Full Framework (Cal.)	3,268	9,884	16,427	26,941
S-Learner	7,563	16,301	24,310	36,033
τV Oracle ($g \equiv 1$)	21,281	34,765	44,615	56,145

At every budget level the calibrated framework matches CATE × LTV within noise (paired t-tests, all $p > 0.05$). The more interesting pattern is that the ranking between value-weighted targeting and CATE Only depends on budget. At tight budgets ($\kappa = 0.10$), CATE Only (4,766) substantially beats CATE × LTV (3,268); at generous budgets ($\kappa = 0.50$) the ordering reverses. The practical recommendation to use $\hat{\tau} \cdot \hat{V}$ therefore applies most strongly at moderate-to-generous budgets. At very tight budgets, pure CATE targeting may be preferable because value weighting dilutes the uplift signal by prioritising high-LTV individuals whose treatment effects may be modest.

A short formal argument underlies the reversal. Write the estimated components as $\hat{\tau} = \tau + \epsilon_\tau$ and $\hat{V} = V + \epsilon_V$ with mean-zero errors independent of τ, V; then

$$\hat{\tau}\hat{V} \;=\; \tau V \;+\; V\,\epsilon_\tau \;+\; \tau\,\epsilon_V \;+\; \epsilon_\tau\,\epsilon_V,$$

so the conditional variance of the product-score at a given (τ, V) is approximately $V^2 \operatorname{Var}(\epsilon_\tau) + \tau^2 \operatorname{Var}(\epsilon_V)$ (dropping the second-order term). The V^2 factor amplifies $\hat{\tau}$-noise most strongly in the right tail of V. Ranking by $\hat{\tau}$ alone escapes this amplification. At tight κ, the top-κ set is drawn entirely from the joint right tail where the amplification is largest, so the product estimator misranks more often than $\hat{\tau}$ alone and the signal gained from including $\hat{V}$ does not offset the extra variance. As κ grows, the top-κ set extends into the body of V where amplification is milder, and the signal carried by $\hat{V}$ (moderate-τ individuals with very high V) dominates; the product estimator then overtakes $\hat{\tau}$. The crossover $\kappa^\star$ depends on $\operatorname{Var}(\epsilon_\tau)$, $\operatorname{Var}(\epsilon_V)$, and the joint tail of (τ, V), and is therefore DGP-specific.

The S-Learner's welfare advantage over the decomposed strategies, first reported at $\kappa = 0.30$ in Table 3, persists across budgets: it is the highest-welfare estimated strategy at every κ.

7.6 Value-Conservatism Frontier

Table 8 and Figure 3 report the welfare-risk trade-off across six configurations of g. All welfare figures use $\mathcal{W}(\pi; \mathbf{1}) = \sum \tau_i \cdot V_i$ on the test set.

No fixed-parameter configuration exceeds the unmodulated welfare of 16,427, and the calibrated configuration matches it exactly by picking $g \approx 1$. One caveat on the frontier itself: in this DGP, risk exposure only varies between 0.136 and 0.149 across configurations, a range of roughly 10%, while welfare varies from 12,588 to 16,427 (about 25%). Changes in g thus move welfare considerably more than they move risk exposure in this simulation. The operator's dial on g is therefore mostly a welfare cost dial, not a dramatic reshaping of who gets treated; in settings where the risk covariate is more concentrated in the treated tail (for example, a population where churn risk clusters sharply), the frontier would show wider risk-exposure variation.

Table 8: Welfare and risk exposure under different g configurations ($N = 10{,}000$, 200 replications, $\kappa = 0.30$). Welfare evaluated under $g \equiv 1$ for every configuration. Labels show $(p_L/p_{ML}/p_{MH})$; all configurations use $p_H = 0.90$ and $\phi = 0.20$ unless stated.

Configuration	Welfare	Risk Exp.	Churn Mod
No risk modulation ($g \equiv 1$)	16,427	0.136	0.440
Wide (0.03/0.50/0.70)	12,588	0.148	0.472
Default (0.05/0.40/0.60)	13,629	0.146	0.468
Narrow (0.10/0.30/0.50)	14,940	0.145	0.464
Aggressive (0.15/0.35/0.50/ϕ=0.05)	13,713	0.149	0.476
Calibrated	16,427	0.136	0.440

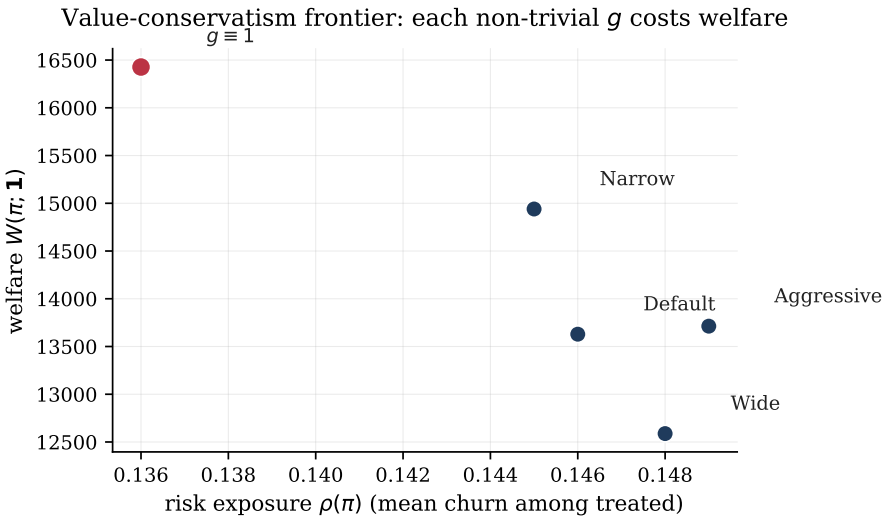


Figure 3: Value-conservatism frontier: each fixed-parameter g sits below the unmodulated $g \equiv 1$ baseline in welfare. The calibrated configuration overlaps the baseline. This is Proposition 3 made visual.

7.7 Robustness

Misspecification of g. We test sensitivity by inverting the risk-modulation function so that it puts high weight at extreme churn levels and low weight in the moderate zone (the opposite of the default). The inverted g achieves welfare 13,905 with risk exposure 0.145, versus 13,629 for the default and 16,427 for $g \equiv 1$. Both non-identity configurations underperform the unmodulated baseline, consistent with Proposition 3.

Alternative CATE estimator. Re-running with an X-Learner (Künzel et al., 2019), which imputes individual treatment effects by plugging each unit's observed outcome and a model-based counterfactual into separate treatment-conditional regressions, produces qualitatively consistent results: CATE Only 20,456, CATE × LTV 20,191, Cal. 20,191. The X-Learner captures the CATE-churn interaction more effectively, so absolute welfare rises, but the qualitative pattern is unchanged.

Depth-matched S-Learner. The published S-Learner uses a deeper tree ensemble (max depth 4) than the decomposed T-Learner (max depth 3), so the welfare comparison reported in Table 3 mixes joint modelling with a hyperparameter difference. We re-run two depth-matched variants on the same 200 seeds: (i) an S-Learner at max depth 3 and (ii) a T-Learner at max depth 4 paired with the existing LTV model. The S-Learner at depth 3 achieves welfare $25{,}213 \pm 4{,}522$ (slightly higher than the published depth-4 S-Learner at 24,310), while the T-Learner at depth 4 paired with LTV achieves $14{,}000 \pm 2{,}738$ (lower than the published depth-3 T-Learner × LTV at 16,427, indicating modest overfitting at the deeper capacity). At common depth 3 the joint-versus-decomposed gap is thus +8,786 welfare units (+53%); at common depth 4 it is +10,310 (+74%). The welfare advantage of joint modelling over the T-Learner-based decomposition in this DGP therefore persists, and modestly widens, when the tree-depth hyperparameter is equalised in either direction.

Correlated-feature DGP. In reality, covariates are correlated (high-spending customers tend to log in more, etc.), so we re-run the simulation with the covariates drawn from a *Gaussian copula* with realistic pairwise correlations ($\rho \in [0.3, 0.4]$). A Gaussian copula is a statistical construction that produces correlated random variables while keeping each individual variable's marginal distribution unchanged, so the individual features still have the shapes specified above but are no longer independent of each other. Under this more realistic design, CATE Only achieves welfare 16,893, CATE × LTV 16,225, and Cal. 16,224, qualitatively identical to the independent-feature results.

Asymmetric-moderation DGP. The main DGP uses a symmetric moderation $m(p) = 4p(1-p)$, which is the most favourable structure for the risk-modulation mechanism and, as Ascarza (2018) notes, unrealistic: real attenuation falls off faster at high churn than at low churn. We therefore re-run the simulation with an asymmetric moderation $m(p) \propto p(1-p)^3$ (peaking near $p = 0.25$ and decaying sharply toward high churn), a structurally different second DGP. The qualitative pattern is unchanged. At $\kappa = 0.30$: CATE Only $36{,}444 \pm 3{,}727$, CATE × LTV $36{,}861 \pm 4{,}322$, Full Framework (Cal.) $36{,}861 \pm 4{,}322$, Full Framework (Est.) $30{,}616 \pm 4{,}297$, Full Framework (Oracle) $54{,}660 \pm 964$, and S-Learner $49{,}991 \pm 4{,}080$. Calibration again selects the near-identity configuration (198 of 200 replications), so Cal. reproduces CATE × LTV; the default fixed g again costs welfare; and the S-Learner again dominates every decomposed estimated strategy. Value weighting now slightly helps rather than slightly hurts at this budget (36,861 vs. 36,444), but the three load-bearing findings (calibration prices rather than discovers g, joint modelling wins, value weighting's sign is budget- and structure-dependent) are robust to the moderation structure. This sign change is itself a deployment hazard worth stating plainly: whether value weighting helps or hurts at a given budget depends on the moderation shape, which the operator cannot observe ex ante. The safe reading is that the value-weighting decision should be made empirically per deployment (via held-out or off-policy evaluation), not

assumed from this paper's main-DGP sign.

Effective (clip-adjusted) welfare. As flagged in Section 7.1, headline welfare is scored on nominal τ despite the outcome clip, which overstates realised lift. Re-scoring the main table on the effective (clip-adjusted) treatment effect $\tau_i^{\text{eff}} = y_{\text{prob},i}^{(1)} - y_{\text{prob},i}^{(0)}$ confirms the bias is modest and, while mildly differential, does not change any ranking: CATE Only 16,634 → 16,044 (3.7% overstatement), CATE × LTV 16,427 → 15,641 (5.0%), Cal. 16,427 → 15,640 (5.0%), Full Framework Oracle 36,641 → 35,180 (4.2%), and S-Learner 24,310 → 22,791 (6.7%). As anticipated, the S-Learner is flattered most, but its effective welfare (22,791) still exceeds the best decomposed strategy's effective welfare (16,044) by a wide margin, so the joint-modelling conclusion is not an artefact of the nominal-welfare convention.

End-to-end joint model (un-handicapped comparator). The S-Learner in Table 3 borrows the decomposed pipeline's value estimate $\hat{V}$, so it is not a fully joint comparator. We therefore also run the end-to-end alternative: a single gradient-boosted model trained to predict the value-weighted outcome ($\hat{V}_i^{\text{proxy}} \cdot Y_i$) directly, scored by the treated-minus-control difference, with no separate value model. It achieves welfare $20{,}158 \pm 6{,}194$, above the best decomposed estimated strategy (CATE × LTV, 16,427) but below the borrowed-$\hat{V}$ S-Learner (24,310). Two things follow. First, joint modelling beats the decomposition even when it does not borrow the value estimate, so the welfare advantage is a property of joint modelling rather than an artefact of the comparison. Second, the borrowed-$\hat{V}$ S-Learner's extra margin over the end-to-end model shows that a well-specified separate value estimate is itself worth something; the strongest estimated strategy in this DGP combines a joint treatment-effect model with a separate value model, which is neither the pure decomposition nor the pure end-to-end joint model.

Diagnostic decomposition (does it finger the right component?). The regret decomposition (Remark 8) claims to identify which component's estimation error drives misranking. We test this directly by computing its two terms on the misranked set $\mathcal{M}$ (the symmetric difference of the oracle and estimated top-κ sets) at $\kappa = 0.30$. Under the well-specified pipeline the CATE-error term $|\tau - \hat{\tau}| \cdot |V|$ averages 81 welfare units on $\mathcal{M}$ against 21 for the value-error term $|\hat{\tau}| \cdot |V - \hat{V}|$, so roughly 80% of near-threshold score error is attributable to treatment-effect estimation, correctly flagging it as the binding component (consistent with the large oracle-to-estimated welfare gap in Table 3). When the CATE model is deliberately degraded (login dropped), the CATE-error term rises to 90 while the value-error term is unchanged at 21: the decomposition localises the added error to exactly the component that was degraded, and leaves the other untouched. This converts the diagnostic from an asserted property into a demonstrated one.

An observational version of the diagnostic. The decomposition above uses oracle τ and V, which a deployed operator lacks. A usable analogue replaces the oracle with a richer cross-fitted *reference* estimator (an X-Learner for τ on all features, a larger ensemble for V) and reads the telescoping terms from the disagreement between the reference and the deployed (T-Learner, misspecified-value) estimates: $|\hat{\tau}^{\text{ref}} - \hat{\tau}| \cdot |\hat{V}|$ and $|\hat{\tau}| \cdot |\hat{V}^{\text{ref}} - \hat{V}|$. This proxy is fully observable. It reaches the same verdict: the CATE-disagreement term (36 units) dominates the value-disagreement term (3), a 93% CATE share, and under CATE misspecification the CATE term rises to 48 while the value term holds at 3 (95% share). The diagnostic is therefore usable in the field, with one honest caveat: disagreement detects *recoverable* error (a dropped feature the reference recovers), not a value-model bias that no available estimator can see; the latter remains invisible to any data-only diagnostic.

Correlation between effect and value. The main DGP makes V roughly independent of τ. Because the value-weighting question turns on whether high-value individuals are also responsive, we inject a controlled dependence and sweep it. At $\kappa = 0.30$, value weighting underperforms pure CATE by a stable 1 to 2% across the range: -1.4% at $\text{corr}(\tau, V) = -0.36$, -2.1% at -0.11 (the

near-independent main case, reproducing the headline numbers), and -2.2% at $+0.20$. Absolute welfare rises with the correlation (from 9,861 to 23,297 for CATE Only) because aligned τ and V raise the achievable total, but value weighting's *relative* verdict at this budget is essentially unchanged, and the S-Learner leads throughout (16,700, 24,599, 32,501). The earlier finding that value weighting does not pay at moderate budgets is thus not an artefact of the independence built into the main DGP.

Heavy-tailed value. Gaming lifetime value is extreme-tailed. Replacing the additive value with a multiplicative log-normal tail (orders-of-magnitude spread, clipped at 100,000) leaves the pattern intact and, if anything, slightly sharpens it: at $\kappa = 0.30$ value weighting underperforms pure CATE by 2.9% (CATE Only 27,360 vs. CATE $\times$ LTV 26,559), consistent with the variance-amplification argument of Section 7.5 (a heavier V tail amplifies $\hat{\tau}$-noise in the product more), and the S-Learner again leads (40,527).

8 Discussion

8.1 Three Scenarios, Three Regimes

The simulation yields a prescriptive guide for practitioners.

- **Well-specified CATE.** Calibration picks $g \approx 1$. Operators who want to exclude extreme-risk segments should choose their preferred g and read the welfare cost off the frontier (Table 8 and Figure 3).
- **Misspecified CATE.** Value-weighted targeting ($\hat{\tau} \cdot \hat{V}$) significantly beats CATE Only (6,955 vs. 6,700). The improvement is from V, not from g.
- **No CATE-risk interaction.** Value-weighted targeting dominates; calibration again picks $g \approx 1$.

The practical decision rule. Use $\hat{\tau} \cdot \hat{V}$ for welfare maximisation at moderate-to-generous budgets; use pure CATE at very tight budgets (Section 7.5). Choose $g \neq 1$ only if the operator has non-welfare preferences to encode, such as risk control, regulatory compliance, or responsible gaming. Run the calibration procedure in either case to price the chosen g against the identity.

Welfare vs. interpretability. The S-Learner beats the best decomposed strategy (CATE Only, 16,634 at this budget) by 46% in this DGP, because a joint model can exploit cross-component interactions the decomposition cannot. The depth-matched ablation of Section 7.7 shows this gap is not an artefact of the tree-depth hyperparameter: at matched depth the gap is $+53\%$ to $+74\%$, so the 46% figure is a lower bound on the joint-modelling advantage in this DGP rather than an upper one. The gap's magnitude is DGP-specific. The decomposed framework buys three things in exchange for this gap: independent auditability, component-wise updating, and transparent operator control via g. Of these, component-wise updating is the one with a direct welfare consequence, and we can demonstrate it. Suppose an operator is running the degraded pipeline of Section 7.3 (CATE model missing login frequency, welfare 6,955) and then improves only the treatment-effect model, restoring the dropped feature, while leaving the value model untouched. Welfare recovers to 16,427, a $+136\%$ gain from a single-component update. A joint model has no analogue: there is no "the CATE part" to re-fit in isolation, so the same improvement requires retraining the whole model end-to-end. This is the concrete operational content of "auditability and component-wise control"; the broader auditability claim (being able to inspect and defend each component separately) is a governance benefit and we do not assign it a welfare number. We also note one benefit the decomposition does *not* buy: robustness under component misspecification. When login frequency is dropped from the CATE features, the decomposed strategies lose 60% of welfare while the S-Learner, a joint model over all features,

is unaffected (Section 7.3); on this axis the joint model is the more robust of the two. In regulated settings such as online gaming, where operators must defend targeting decisions and different teams maintain different models, the auditability exchange can still be worthwhile, but it should be made on governance grounds, with the substantial welfare cost (and the absence of a robustness advantage) stated plainly rather than assumed away.

8.2 Estimation in Practice

The composite score needs three estimates: $\hat{\tau}(x)$, $\hat{V}(x)$, and $\hat{R}(x)$. The framework is agnostic to the choice of estimator. Proposition 2 shows that errors in components with large magnitude, particularly V (which varies by orders of magnitude), are amplified by the other components and so contribute disproportionately to welfare loss.

Calibrating the parameters of g is essential. The default parameterisation costs 17% of welfare; calibration recovers the gap. For deployment we recommend the held-out validation procedure from Section 7.1, with the identity configuration included in the candidate grid.

The calibration procedure evaluates candidates against estimated welfare $\mathcal{W}(\hat{\pi}; \mathbf{1})$. Because Proposition 3 guarantees that $g \equiv 1$ maximises unrestricted welfare, any welfare-based calibration will pick $g \approx 1$ by construction. This is a consistency check on the theory, not a new finding. The primary practical role of calibration is therefore cost quantification. Each candidate g on the validation fold produces a point on the frontier; the operator reads off the welfare cost of their preferred preference configuration and decides whether the targeting control is worth it. The operator sets g from domain preferences; calibration tells them what g costs.

A natural follow-up question is whether a different calibration objective could allow the procedure to discover preference-useful g rather than always returning the identity. Candidates include a penalty-augmented welfare objective (estimated welfare minus an operator-supplied penalty on deviations from a preferred risk profile), a minimax-regret objective (selecting g that minimises the worst-case welfare loss across a plausible set of CATE misspecifications), or a constrained objective that caps risk exposure at an operator-chosen ceiling. Each of these changes the calibration problem from "find the welfare-optimal g" (which Proposition 3 already answers) into "find the best trade-off against a separately specified criterion", allowing non-identity solutions to arise. We do not pursue these here because they require additional operator input beyond the scope of the present framework, but they are natural extensions of the calibration machinery.

Remark 12 (Churn proxy vs. structural churn probability)**.** *The churn model in the simulation is trained to predict $1 - Y_i \mid W_i = 0$, the control-group non-retention indicator. This estimates $P(Y(0) = 0 \mid X)$, which is correlated with but not identical to the structural churn probability p_i^{churn} generated from the logistic model in the DGP. The distinction matters because the risk-modulation function g operates on estimated churn: if the churn model captures different information than structural churn risk, the effective modulation function differs from the intended one. In practice, churn models are typically richer than the binary proxy used here, incorporating longitudinal behavioural signals; the simulation understates both the difficulty and the potential value of churn estimation.*

8.3 Evaluation in Practice

The simulation evaluates strategies using oracle (ground-truth) potential outcomes. In deployment those are never available: we only ever observe one of the two outcomes per individual (treated or not), which means realised welfare has to be *estimated*, not measured directly. The standard approach is *off-policy evaluation*: estimating what welfare a proposed policy would have produced using data collected under a different (typically randomised) policy. Two families of estimators are suitable. The first is the *augmented inverse propensity weighting* (AIPW)

estimator (Dudik et al., 2011), which combines a model of the outcome with the inverse of the treatment-assignment probability (the *propensity score*). AIPW is *doubly robust*: it produces a valid welfare estimate if either the outcome model or the propensity model is correctly specified (both can be wrong, but only one at a time). The second is *counterfactual risk minimisation* (Swaminathan and Joachims, 2015), which reframes policy learning as minimising an importance-weighted version of the empirical loss on historical data ("importance-weighted" meaning each observation is reweighted by how likely the proposed policy would be to assign its observed treatment, relative to the policy that actually collected the data). The treatment-effect estimates already used inside the composite score can double as the outcome model in AIPW, so the targeting and evaluation pipelines share component estimates. In deployment we recommend AIPW-based welfare monitoring combined with a randomised holdout group that receives random treatment assignments, to maintain valid propensity scores over time.

8.4 Assumptions and Scope

The multiplicative welfare function (Definition 3) requires that the operator's valuation factor into the product of treatment effect, outcome value, and risk modulation (it is *separable* in the three components). We adopt separability deliberately, for two reasons, and we are explicit that our optimality results are conditional on it rather than derived from more primitive axioms. First, separability is the form the heuristic the paper is about already takes: practitioners rank by $\hat{\tau} \cdot \hat{V}$, a product, so the multiplicative welfare function is the objective implicitly revealed by existing practice, not an arbitrary imposition. Second, separability is exactly what buys the decomposition, auditability, and component-wise update properties that distinguish this framework from a single joint model; a non-separable objective would not merely complicate the algebra, it would dissolve the framework's reason to exist. The honest reading is therefore that separability is a scope condition: where it holds, the composite-score rule is welfare-optimal and the regret decomposition applies; where it fails, the framework should not be used. This rules out non-separable preferences. As a concrete example: suppose an operator's willingness to take risk exposure varies with customer value; they are more willing to target stable (low-churn) high-value customers than stable low-value customers, because the returns when retention goes through are larger in the former case. The effective preference weight then depends jointly on R and V, so the preference factor becomes $g(R, V)$ rather than $g(R)$, and the composite score no longer factors into a product of three univariate components. Such preferences require a more general welfare function than the one used here, and the decomposition, auditability, and component-wise update properties that motivated our framework would weaken accordingly; an operator who holds them should treat that as a signal to abandon the decomposed framework rather than to patch it. The binary outcome assumption ($Y_i \in \{0, 1\}$) restricts the framework to the *extensive margin*: the outcome is whether a thing happened (the customer retained, the patient recovered), not how much. Extension to continuous outcomes ("how much revenue did the customer generate in the post-treatment window") is straightforward and does not change any of the results. With $Y \in \mathbb{R}$, the CATE $\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$ is simply the expected continuous-valued gain, and V may be redundant (or may be interpreted as a multiplier that re-weights the continuous outcome to a common welfare unit, e.g., monetising QALYs). The composite score $\tau(x) \cdot V(x) \cdot g(R(x))$ and all three propositions go through with obvious modifications. In many operational gaming settings the continuous outcome (expected value over a retention window) is the quantity operators actually optimise; the binary framing here is chosen for clarity of exposition and because retention is the standard reference outcome in the uplift literature. The pre-treatment requirement for the auxiliary components (Remark 2) is a real restriction. In sequential deployment, both churn probability and outcome value at a later period will reflect the effects of prior treatments, which means they are no longer strictly pre-treatment quantities; this introduces a statistical problem called *endogeneity* (the auxiliary variables become correlated with the unobserved drivers of the new treatment's effect), and the

causal interpretation of the composite score breaks down. We assume treatment is equally costly for every individual; if costs differ (e.g., bonuses of different sizes), the appropriate ranking is by the net-benefit score $s_i - c_i$ rather than s_i, where c_i is the per-individual treatment cost.

The SUTVA assumption (Assumption 1(iii)), no interference between individuals, may be violated in settings where individuals interact: gaming platforms with guilds, multiplayer modes, or shared play sessions; clinical trials where patients influence each other; online services with social features. If treating individual i affects the outcome of individual j, the individual-level treatment effect τ_i is no longer well-defined: it depends on whether i's peers are also treated. This kind of spillover is known in the causal-inference literature as *network interference* (each individual's outcome depends on the treatment assignments of their network neighbours, not just their own treatment). Extending the framework to account for network interference is a direction for future work.

Remark 13 (Terminology: risk modulation vs. risk aversion)**.** *The "risk-modulation" function g encodes covariate-dependent targeting weights based on an observable auxiliary variable (churn probability), not risk aversion in the decision-theoretic sense of Manski and Tetenov (2007). The function g is better understood as a domain-informed reweighting of the welfare integrand.*

The multiplicative CATE decomposition $\tau(x) = \tau_0(x) \cdot m(R(x))$ assumed in Remark 11 is an approximation. Additive interactions and threshold effects are not well-served by the multiplicative framework; the operator should assess whether the domain structure supports multiplicative moderation before deploying g.

8.5 Extensions

Natural directions for future work include: (i) multi-armed and sequential treatment with bonus sizing (selecting from a portfolio of intervention types); (ii) finite-sample regret bounds for the three-component product score analogous to the $O(n^{-1/2})$ rates of Athey and Wager (2021); and (iii) fairness constraints within the budget-constrained optimisation.

9 Conclusion

Proposition 1 places value-weighted uplift targeting inside the budget-constrained empirical-welfare-maximisation framework: under the multiplicative-separable welfare function of Definition 3 and a budget constraint, the welfare-maximising deterministic rule ranks by $\tau \cdot V$ (or, with operator preferences, by $\tau \cdot V \cdot g(R)$). The decision-theoretic merit of value weighting is established in prior marketing work; the multiplicative separability that lets it nest into policy learning is itself a modelling assumption, not a derivation, and its scope is discussed in Section 8. The paper's two more durable contributions are the regret decomposition and the operating-regime map it explains. The regret decomposition (Proposition 2, Remark 8) localises welfare loss to individuals misranked near the threshold and attributes it to a specific component's estimation error, so it tells an operator which of the three estimates to improve first. The operating-regime map is the simulation's main empirical finding: value weighting matters most exactly where it is most needed. Under CATE misspecification, $\hat{\tau} \cdot \hat{V}$ significantly beats CATE Only (6,955 vs. 6,700) because the value model retains features the uplift model drops. The effect's direction is structural; its magnitude is DGP-specific. Value weighting's advantage is also budget-dependent: at moderate-to-generous budgets it matches or exceeds CATE Only, but at very tight budgets pure CATE is stronger because value weighting dilutes the uplift signal.

Adding the operator-configurable preference weight g extends the framework to encode domain preferences (excluding extreme-risk segments, responsible-gaming considerations, regulatory constraints). Proposition 3 prices this: any $g \neq 1$ weakly reduces welfare relative to $g \equiv 1$. The

simulation's calibration procedure reproduces the identity in every scenario, which is what Proposition 3 implies it must do; the substantive role of calibration is to produce the welfare-cost schedule across different preference configurations. The operator's role is not to discover a welfare-improving g (there is none, by construction); it is to decide which preferences are worth what welfare cost.

The S-Learner beats every decomposed strategy on welfare in this DGP. The depth-matched ablation, the end-to-end joint model, and the effective-welfare re-scoring (all in Section 7.7) confirm this gap is not an artefact of tree depth, the borrowed value estimate, or the nominal-welfare convention. An end-to-end joint value-weighted model (Section 7.7) confirms this is a property of joint modelling, not of the particular S-Learner that borrows the decomposed pipeline's value estimate. When a single flexible model is available and interpretability is not required, direct policy learning wins on welfare. The decomposed framework earns its keep where transparency, auditability, and component-wise control matter, such as regulated online gaming, where the piecewise linear g gives operators five interpretable knobs with operational meanings.

We close by stating the paper's posture plainly, because it is as much a caution as a method. An operator who optimises welfare alone should, on this evidence, reach for a single joint model and not for the decomposed framework; the operator-preference weight g should be set from domain preferences and priced, never tuned for welfare, because it provably cannot improve it; and value weighting should be switched on at moderate-to-generous budgets and off at tight ones. What the framework offers is not a higher-welfare estimator but a common objective in which these statements can be made precisely, a regret decomposition that says which estimate to fix first, and a transparent, auditable structure whose welfare cost is now quantified rather than assumed. Whether that transparency is worth its welfare cost is a governance judgement we make explicit rather than resolve.

The main limitation is the absence of real-data validation: the qualitative findings are stable across the structural variations we test (two moderation shapes, a correlation sweep between effect and value, and a heavy-tailed value distribution), but deployment performance ultimately depends on the quality of the component estimates in the field. For production deployment, the framework should be paired with off-policy welfare evaluation (e.g., AIPW estimators) on observed data. Future work should validate on real targeting data, extend to multi-armed and sequential treatment, and derive finite-sample regret bounds for the three-component product score.

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Conflict of Interest and Disclosure

The author is employed by Allwyn, an operator in the gaming and lottery sector, and the framework studied here is directly relevant to that sector. This is a potential conflict of interest and is disclosed in full. The views expressed are the author's own and do not necessarily represent the position of Allwyn. The research received no specific external grant funding. The framework is presented with explicit responsible-gaming safeguards (Section 6.2), and the author draws an explicit distinction between operator welfare (the econometric quantity the method optimises) and player welfare, cautioning that welfare-optimal targeting in the econometric sense is not the same as ethically optimal targeting in a harm-sensitive domain.

Ethics Statement

This study involves no human participants, no personal data, and no experiments on individuals: all results are produced by a synthetic, fully specified simulation (Section 7), and no real player, customer, or patient data was used. No ethics-board approval was therefore required. Because the application domain (promotional targeting in online gambling) carries real potential for harm, the paper foregrounds that risk rather than abstracting it away: it documents that value-weighted targeting can concentrate incentives on the most-harmed players (Section 6.2), treats a vulnerability-based exclusion set and a harm gate as preconditions for any responsible deployment, recommends conservative defaults (a zero floor on the highest-risk segment), and notes that direct promotion to identified at-risk players may itself be restricted in some jurisdictions. The framework should not be deployed in a gambling setting without the responsible-gaming safeguards described in Section 6.2 and the validated harm-measurement and regulatory infrastructure they depend on.

Data and Code Availability

This study uses no empirical data; all reported quantities are generated by simulation. The complete simulation code, the revision-additions code (asymmetric-moderation and correlated and heavy-tailed value DGPs, the harm-gate and hard-constraint analyses, the component-update and end-to-end-joint comparisons, the clip-adjusted welfare re-evaluation, and the observational diagnostic), the data-generating process specification, the fixed random seeds ($42 + 7r$ for replication r), and the output files needed to reproduce every table and figure are available in the companion repository. The entire study is deterministic and reproducible from those scripts.

A The Piecewise Linear Risk-Modulation Function

This appendix gives the formal definition of the five-parameter risk-modulation function g used in Section 6 and its regularity properties. The functional form is an implementation choice; the operational reading of the parameters and the default values θ_0 are in the body.

Definition 5 (Piecewise linear risk-modulation function)**.** *The risk-modulation function* $g : [0,1] \to [0,1]$ *is defined as*

$$g(p;\,\theta) = \begin{cases} 0 & \text{if } p < p_L, \\ \dfrac{p - p_L}{p_{ML} - p_L} & \text{if } p_L \le p < p_{ML}, \\ 1 & \text{if } p_{ML} \le p \le p_{MH}, \\ 1 - (1-\phi)\,\dfrac{p - p_{MH}}{p_H - p_{MH}} & \text{if } p_{MH} < p < p_H, \\ \phi & \text{if } p \ge p_H, \end{cases} \tag{19}$$

parameterised by $\theta = (p_L, p_{ML}, p_{MH}, p_H, \phi)$ *with* $0 \le p_L < p_{ML} \le p_{MH} < p_H \le 1$ *and* $\phi \in [0,1]$.

Remark 14 (Properties of the piecewise linear g)**.** *The function* g *defined in* (19) *satisfies: (i)* g *is continuous on* $[0,1]$*; (ii)* g *is non-decreasing on* $[0, p_{MH}]$ *and non-increasing on* $[p_{MH}, 1]$*; (iii)* $g(p) = 0$ *for* $p < p_L$ *and* $g(p) \ge \phi$ *for all* $p \ge p_H$*; on the ascending ramp* $[p_L, p_{ML})$*,* $g(p)$ *can be below* ϕ *when* $\phi > 0$*; (iv)* g *is* Lipschitz continuous *(its rate of change is bounded by a constant) with constant* $L = \max(\frac{1}{p_{ML} - p_L}, \frac{1-\phi}{p_H - p_{MH}})$*; intuitively, the steeper of the two ramps determines how fast* g *can change. Standard results in approximation theory guarantee that any Lipschitz risk-modulation function can be approximated to within an error of* ϵ *by a piecewise linear function with the number of breakpoints growing proportionally to* $1/\epsilon$ *(so halving the*

allowed error roughly doubles the number of breakpoints needed). Five breakpoints, as here, therefore already buy a substantial amount of shape flexibility.

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